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Physics–Semantic Consistency-Constrained Association of Heterogeneous RF Observations for Multi-Receiver Communication Signal Monitoring

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Abstract

We study semantic alignment of heterogeneous RF receiver reports whose native class-probability columns may have different meanings and partly missing correspondences. A known reference vocabulary fixes the common class names, while physical report groups provide evidence for the missing maps. Receiver-pair score tables support exact conditional search within a small vocabulary. The recovered maps then support event association under receiver exclusivity and pairwise physical gates. We compare reference-only fitting, coordinate updates, and an adapted spectral synchronization method. Independent four-class tests recover 120/120 hidden entries across ten map interventions. Five adaptation scenes per intervention restore event-pair F1 to 0.7997 with unweighted linear scoring and 0.8232 with boosted scoring. Spectral synchronization gives comparable results, and alternating event updates provide no additional accuracy gain in the tested settings. Dense same-class collisions reduce event F1 to about 0.45 despite complete map recovery. Shared wrong class predictions can preserve event compatibility, showing that class registration, class correctness, and event identity are distinct properties. An auxiliary two-gateway LoED check gives reference-packet F1 of 0.9975. Clock fitting from training metadata reproduces that result, although packet-reference ambiguity and remaining supervision limit its interpretation. Validation of the class-probability mechanism relies on controlled descriptors.

Keywords: semantic alignment; heterogeneous RF observations; multi-sensor fusion; signal reconnaissance; physical consistency

1. Introduction

A signal-reconnaissance system combines reports of detected activity from several RF receivers. A recognizer may also supply probabilities over signal families [1–3]. Differences in units, class orders, and measurement errors create multi-source heterogeneity. All sources in this study are RF receivers.

Here, *semantic evidence* is probability compatibility after class columns have been mapped to common meanings; it does not refer to natural-language interpretation or decoded message content. For example, column 1 can denote different families at two receivers, so comparing those columns directly can produce false agreement. Correct mapping makes the probabilities comparable, but separate transmissions can still share a family. Event identity therefore requires a further inference step.

We distinguish three alignment levels. Representation alignment converts values and uncertainties to common units. Class-semantic alignment registers native probability columns to a shared signal-family vocabulary. Instance alignment groups reports of one bounded transmission. The title's *consistency* refers to the resulting groups: membership is transitive, each receiver appears at most once per group, and every pair passes the physical gate. These requirements enforce the input contract without certifying event identity.

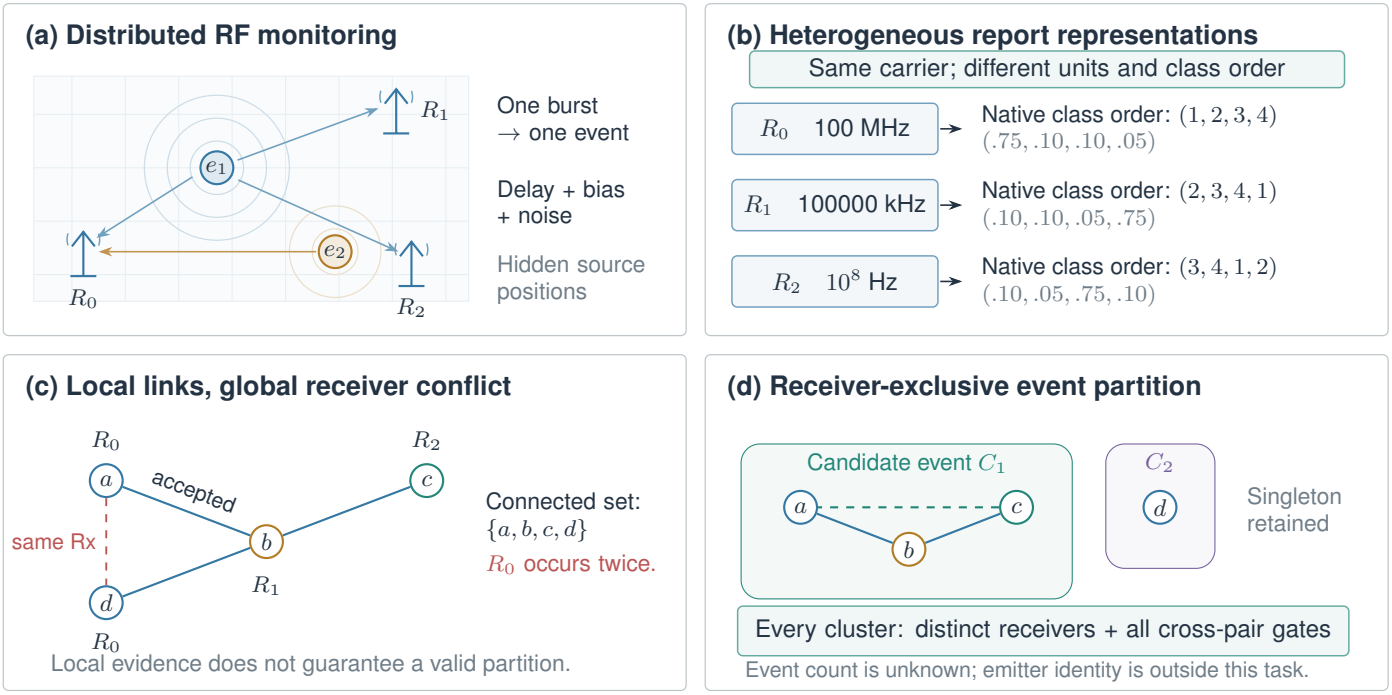


Figure 1. The alignment task. Native units and class orders differ across RF receivers. Hidden source positions illustrate the generating scene. They are not estimated or supplied to inference. A receiver-exclusive partition is a task requirement. It is not an observed accuracy result. A valid group also requires every pair to pass physical admission. The panels are conceptual examples.

The source registry states field meanings, units, and uncertainty declarations [4–6]. The baseline also supplies class maps, while the extension estimates selected missing permutations within a known four-class vocabulary. One reference receiver retains its known map to fix the otherwise arbitrary common class naming. The method does not learn field definitions or new classes.

Unit conversion, calibrated scoring, assignment, and signed clustering are established components [7–10]. Permutation synchronization also estimates several maps jointly [11,12]. Here, class correspondences must be recovered from uncertain RF reports whose event groups are initially unknown. We address this report-level alignment problem and test its limits:

1. We specify an auditable RF observation contract. It separates representation, class-semantic, and instance alignment. Matched unit and vocabulary re-encodings leave canonical evidence and event partitions unchanged.
2. We formulate missing class maps and event groups in one score model. Receiver-pair tables retain every candidate relative map. Exact conditional search provides a small-problem reference. We compare it with coordinate updates, reference-only fitting, and an adapted spectral synchronization method. A conditional identification argument states the role of a known reference and informative pair scores.
3. We test the alignment across frozen scorers and independent descriptor scenes. Same-class collisions and classifier-error interventions expose the limits of semantic evidence.

Integer programs measure the cost of approximate event grouping. Public records provide a separate check of metadata correspondence and clock preprocessing.

The inference target is a bounded transmission. Receiver locations are known, while source locations and event count are hidden. The method starts after detection and does not evaluate a waveform front end [13–15].

Controlled descriptors isolate class-map and association effects. We retain the initial 800-scene benchmark for continuity. Separate adaptation and test scenes support the mapping comparisons and semantic stress tests. Public LoED records provide an auxiliary metadata check [16,17]. We use them to test reference-packet correspondence and preprocessing sensitivity. The controlled benchmark supplies the evidence for the class-probability mechanism.

2. Related Work

2.1. RF measurements and semantic interoperability

RF recognition estimates signal families. Its reliability depends on channel conditions, representations, and datasets [1–3,18,19]. Our inputs retain class probabilities, but many transmissions can share a family, so the family label cannot uniquely identify an event.

Differential-delay estimation and hyperbolic localization use timing information [20, 21]. Recent work examines localization trustworthiness, synchronization, and carrier offsets [13,22,23]. We use reported descriptors and a necessary pair-delay bound based on receiver separation, without estimating source coordinates. Pairwise feasibility does not necessarily imply a common feasible source position.

SSN and SOSA describe sensors, procedures, and observed properties. PROV-O describes information origin [4,5,24]. Interoperability research addresses incompatible formats and vocabularies [6,25]. Our registry follows this separation of value and meaning. The extension learns missing permutations within a declared vocabulary. It does not learn an ontology or reconcile class hierarchies.

Emitter recognition estimates persistent transmitter identity. WiSig and collaborative RF-fingerprint studies examine receiver and channel variation [26–29]. Domain adaptation addresses related shifts [30–32]. One transmitter can produce many events; our correspondence unit is a single transmission, and the inputs contain no waveform fingerprints.

2.2. Association, clustering, and confidence

Assignment, multiple-hypothesis tracking, and probabilistic association estimate measurement origin [14,33]. Classification-aided inference combines class and spatial evidence [34]. Random finite-set models represent object count and trajectories [35]. Passive and multistatic systems impose further correspondence structure [15,36,37]. These methods establish the need to assess whole assignments. They do not supply the observation contract for this RF report benchmark.

The Hungarian method solves two-sided assignment [8]. Merging separate receiver-pair assignments can create globally conflicting groups. Signed correlation clustering instead scores a partition directly [9]. Pruned-pivot methods address scaling, while min-max clustering uses a different loss [38,39]. Must-link and cannot-link methods impose pair restrictions [10]. Our objective belongs to this established family, with receiver exclusivity and physical admission as RF-specific restrictions. We evaluate their accuracy–feasibility tradeoff without claiming a general improvement over these method families.

Temperature scaling adjusts numerical confidence [7]. Calibration reviews distinguish probability assessment from ranking [40,41]. Distribution shifts can damage calibration [42–44]. We assess pair probabilities separately from partitions. The partition objective is

Table 1. Relation to other methods. The table lists what each method estimates and how it relates to this study. It does not claim that all earlier methods lack global consistency.

Method family	Primary inference target	Role and boundary in this study
RF sensing and recognition	Activity or signal class	Upstream descriptors, not common-event labels [1,2].
Physical localization	Delay or source position	Necessary geometric test, not a waveform or localization benchmark [13,20,21,23].
Multi-sensor association	Measurement origin or trajectory	Structural precedent, not directly comparable tracking performance [14,15,33].
Sensor semantics	Meaning and provenance	Declared field meanings, partly learned class permutations, and event correspondence; no ontology learning [4–6,24].
Permutation synchronization	Compatible maps across sources	Structural precedent for class-map recovery; adapted spectral baseline uses the same provisional RF groups [11,12].
Constrained clustering	Unknown-count partition	Signed objective with receiver and gate requirements [9,10].

a composite score. Shared receiver errors prevent an interpretation as independent pair likelihoods.

2.3. Class-map completion and permutation synchronization

Permutation synchronization jointly reconciles pairwise maps [11]. MatchLift estimates consistent partial correspondences from noisy pairwise matches [12]. These studies already establish global map consistency as an inference problem. Joint search and receiver-pair factorization alone therefore do not establish a new general matching algorithm.

Our map variables register native signal-family columns, while the event variables group individual transmissions; one class can occur in many events. Provisional RF event groups supply report pairs whose probabilities define a score for every relative class permutation. Exact conditional search retains the full score tables. The adapted spectral comparator selects one relative permutation per receiver pair and then synchronizes those choices, discarding score margins between candidate maps. Both methods use the same provisional groups and pair evidence. Their comparison tests the value and cost of retaining the full tables in this RF formulation. It does not reproduce every setting or guarantee of the original synchronization paper.

Precision–recall and partition indices have different denominators [45,46]. We report both accuracy and feasibility. Scene-level splits prevent shared events or receiver residuals from crossing training and test sets [47]. Uncertainty intervals use refitted replications rather than individual report pairs.

3. Methodology

Figure 2 shows how registry-based interpretation and physical–semantic scoring produce aligned records and event groups. Training and calibration are separate offline operations: scoring uses *all cross-receiver training pairs*, whereas temperature fitting uses admitted calibration pairs only.

3.1. Semantic alignment scope and observation contract

Let o_i^{nat} denote the native report from source s_i. The declared registry $\mathcal{A}_s$ specifies field meanings, unit scales, class correspondences, and the source of each uncertainty declaration. Applying it gives a common observation record:

$$\bar{o}_i = \mathcal{A}_{s_i}(o_i^{\mathrm{nat}}), \qquad \mathcal{Y} = \{(\bar{o}_i, s_i, z_i) : i = 1, \ldots, N\}, \tag{1}$$

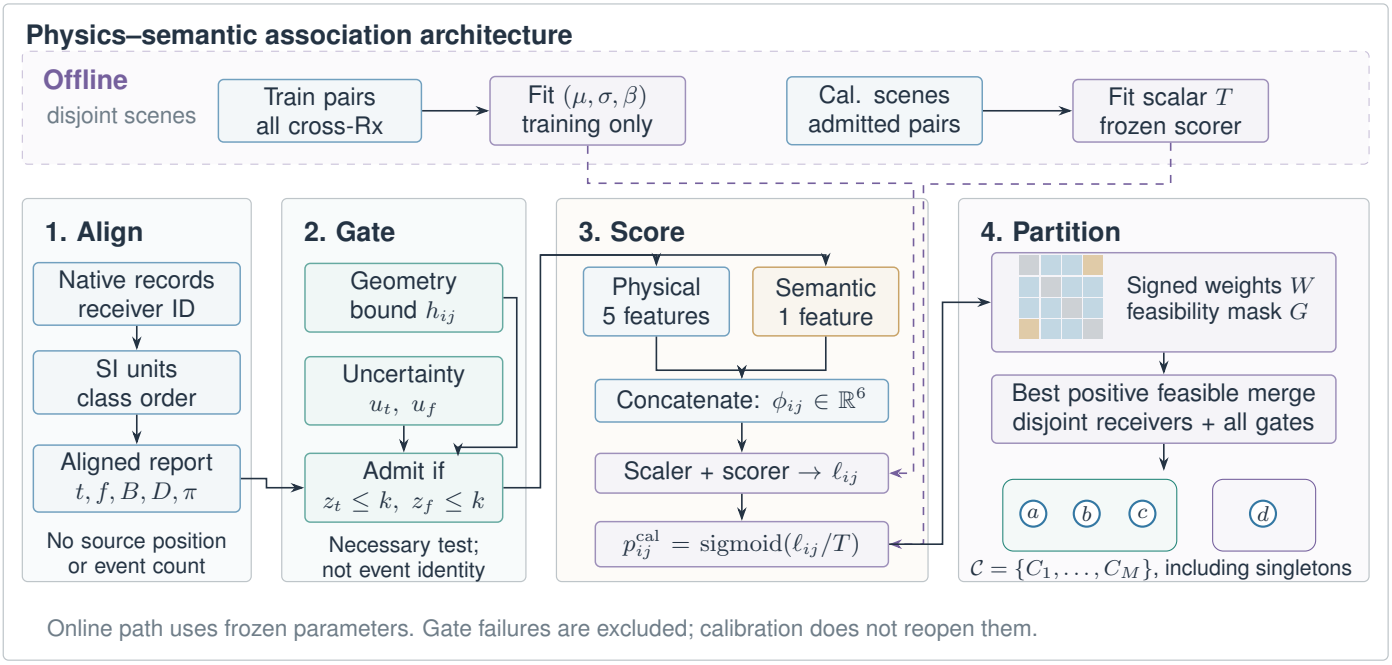


Figure 2. Processing and fitted parameters. Solid arrows carry reports or evidence; dashed arrows supply frozen parameters. Scoring uses all training pairs, and temperature fitting uses admitted pairs from separate calibration scenes. The affinity sketch is illustrative. Rejected pairs do not enter decoding, which receives neither source coordinates nor event count. Figures 3 and 4 detail admission and decoding; Supplementary Figures S3–S5 detail representation, evidence, and calibration.

The comparable record $\bar{o}_i$ retains source identity, while z_i denotes its inferred instance. A decision $z_i = z_j$ assigns one transmission without equating the receivers' measurements or class predictions.

Field meanings, units, uncertainties, vocabulary size, and a reference map are supplied; disclosed anchors are assumed correct. Section 3.7 infers selected missing permutations together with event groups. Many-to-one, hierarchical, and open-set maps are unsupported and rejected.

3.1.1. RF observation model

Consider S receivers at known positions $\mathbf{r}_s \in \mathbb{R}^2$ (m), indexed by $s \in \{0, \ldots, S-1\}$. A window contains N reports $i \in \{1, \ldots, N\}$. Event e has hidden time τ_e (s), position $\mathbf{x}_e$ (m), carrier f_e (Hz), bandwidth B_e (Hz), duration D_e (s), and family $q_e \in \{1, \ldots, K\}$. The number of visible events is unknown. A non-clutter report from s_i follows

$$t_i = \tau_e + \|\mathbf{x}_e - \mathbf{r}_{s_i}\|/c + b_{s_i} + \epsilon_i^t + m_i, \tag{2}$$

$$f_i = f_e + d_{s_i} + \epsilon_i^f, \qquad B_i = B_e + \epsilon_i^B, \qquad D_i = D_e + \epsilon_i^D. \tag{3}$$

We assume propagation speed $c = 299{,}792{,}458\ \text{m s}^{-1}$. Clock offset b_s (s) and oscillator offset d_s (Hz) are shared within a receiver's scene, without test-truth correction. The ϵ terms are independent, zero-mean measurement errors. Additional delay $m_i \ge 0$ (s) is zero outside its stress test. Bandwidth and duration are clipped positive; the generator produces descriptors rather than waveforms.

Receiver exclusivity requires consolidation to at most one report per transmission, with missed detections allowed. Independent clutter reports receive singleton labels; a common false interferer is not modeled. Inference uses receiver identities and positions, measurements, declared errors, native units, class maps, and probabilities. Event labels, source positions, and clutter flags remain evaluation-only.

3.2. Representation and class-semantic alignment

A native carrier value is v_i, and its unit is u_i. We convert it to the common unit as follows:

$$f_i = \alpha(u_i)v_i, \qquad \alpha(\mathrm{Hz}) = 1, \quad \alpha(\mathrm{kHz}) = 10^3, \quad \alpha(\mathrm{MHz}) = 10^6. \tag{4}$$

Time and duration use seconds and bandwidth uses hertz, with declared property meanings. Matrix M_s registers native columns: (q, a) is one when column a denotes common family q. Thus

$$\pi_i = M_{s_i}\pi_i^{\mathrm{nat}}, \qquad \pi_i \in [0,1]^K, \quad \sum_q \pi_{iq} = 1. \tag{5}$$

Missing evidence is uniform. Mapping resolves column meanings but supplies no new evidence, unknown classes, or finer concepts. The generator uses cyclic permutations; the audit also tests arbitrary receiver-specific orders with matching registries. Supplementary Figure S3 illustrates units, class meanings, and provenance.

Representation consistency.

For invertible re-encoding R_s, a matched registry $\mathcal{A}'_s = \mathcal{A}_s \circ R_s^{-1}$ gives $\mathcal{A}'_s(R_s o_i^{\mathrm{nat}}) = \bar{o}_i$. Features, gate decisions, and deterministic partitions are then unchanged up to numerical precision.

This establishes consistency under a known registry, not unknown-map recovery. A shared class renaming preserves inner products, while unmatched source-specific permutations generally change them.

Declared measurement deviations are $\sigma_i^t > 0$ and $\sigma_i^f > 0$. Receiver offsets $\mathbf{b} = (b_0, \ldots, b_{S-1})^{\mathsf{T}}$ and $\mathbf{d} = (d_0, \ldots, d_{S-1})^{\mathsf{T}}$ have covariance matrices Σ_b and Σ_d, in s^2 and Hz^2. With measurement errors independent of these offsets,

$$\begin{aligned} u_{t,ij}^2 &= (\sigma_i^t)^2 + (\sigma_j^t)^2 \\ &\quad + [\Sigma_b]_{s_i,s_i} + [\Sigma_b]_{s_j,s_j} - 2[\Sigma_b]_{s_i,s_j}, \end{aligned} \tag{6}$$

$$\begin{aligned} u_{f,ij}^2 &= (\sigma_i^f)^2 + (\sigma_j^f)^2 \\ &\quad + [\Sigma_d]_{s_i,s_i} + [\Sigma_d]_{s_j,s_j} - 2[\Sigma_d]_{s_i,s_j}. \end{aligned} \tag{7}$$

Entries $[\Sigma_b]_{s,t} = \mathrm{Cov}(b_s, b_t)$ and $[\Sigma_d]_{s,t} = \mathrm{Cov}(d_s, d_t)$ use row and column indices; each cross-covariance is subtracted twice. Positive square roots define $u_{t,ij}$ and $u_{f,ij}$. Nominal off-diagonal covariances vanish, but shared receivers still couple pairs. Correlation tests compare diagonal and matched declarations. Bandwidth and duration use $u_{B,ij}^2 = (\sigma_i^B)^2 + (\sigma_j^B)^2$ and $u_{D,ij}^2 = (\sigma_i^D)^2 + (\sigma_j^D)^2$. These are declared inputs, not estimates from held-out labels.

3.3. Physical admission and its conditional coverage

For reports from distinct receivers, set $\Delta t_{ij} = t_i - t_j$ and $\Delta f_{ij} = f_i - f_j$. The receiver distance gives the following delay bound:

$$h_{ij} = \frac{\|\mathbf{r}_{s_i} - \mathbf{r}_{s_j}\|}{c}, \qquad \frac{\big|\|\mathbf{x}_e - \mathbf{r}_{s_i}\| - \|\mathbf{x}_e - \mathbf{r}_{s_j}\|\big|}{c} \le h_{ij}. \tag{8}$$

The reverse triangle inequality requires no source estimate and motivates the normalized residuals and gate:

$$z_{t,ij} = \frac{\max(0, |\Delta t_{ij}| - h_{ij})}{u_{t,ij}}, \qquad z_{f,ij} = \frac{|\Delta f_{ij}|}{u_{f,ij}}, \tag{9}$$

$$g_{ij}(k) = \mathbf{1}[z_{t,ij} \le k]\mathbf{1}[z_{f,ij} \le k], \qquad k > 0. \tag{10}$$

The indicator $\mathbf{1}[A]$ equals one when A holds and zero otherwise; g_{ij} combines the two tests. Figure 3 shows uncertainty widening the delay interval. Time and frequency determine admission; bandwidth and duration enter scoring. Rejection means exceeding the declared tolerance, not physical impossibility.

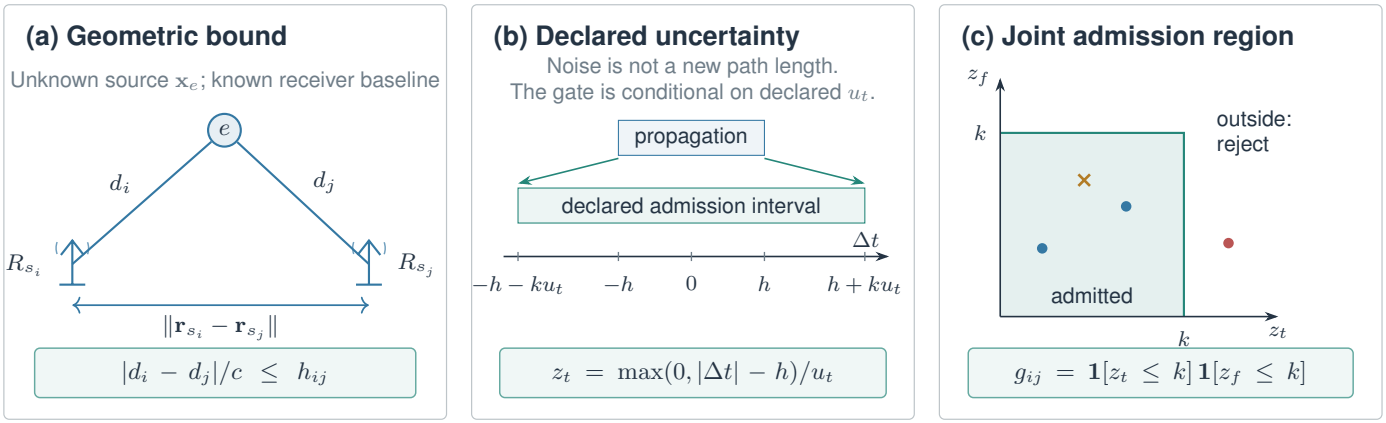


Figure 3. Physical admission. (a) Path lengths $d_i = \|\mathbf{x}_e - \mathbf{r}_{s_i}\|$ and d_j obey the receiver-baseline bound. (b) Uncertainty expands $[-h, h]$ to $[-h - ku_t, h + ku_t]$. (c) Both residuals must pass. Circles represent possible true pairs and the cross a false pair: either type can pass or fail. These are illustrations, not measurements.

Proposition 1 (Conditional true-pair rejection bound)**.** *Consider a same-event pair with $m_i = m_j = 0$. Assume the pair's timing error is zero-mean Gaussian with the declared variance. Assume the same for its frequency error. Assume also that the source carrier follows Equation (3). Then*

$$\Pr\{g_{ij}(k) = 0 \mid y_{ij} = 1\} \le 4\Phi(-k), \tag{11}$$

where $y_{ij} = 1$ denotes a true same-event pair and Φ is the standard-normal cumulative distribution function.

Proof is given in Supplementary Section S10. The nominal $k = 3.5$ gives Gaussian rejection bound 9.31×10^{-4}. This tail budget is invalidated by underestimated bias, mean offsets, or excess delay and is not a universal NLOS setting. Pairwise admission also does not establish a common source position.

3.4. Physical–semantic evidence for instance alignment

What the semantic variable represents.

The latent family is q_e; the reported belief in family q is π_{iq} for observation i. A deployed recognizer requires a documented vocabulary and method. Here $K = 4$ anonymous labels $Q_1, \ldots, Q_4$ are generated with the descriptors, without waveform classification or assigned WiFi/LoRa protocol names.

Vectors $(0.7, 0.1, 0.1, 0.1)$ and $(0.6, 0.2, 0.1, 0.1)$ favor the same family but can describe different transmissions. We assess whether such evidence helps association, rather than measuring protocol recognition.

For a report pair, set $\Delta B_{ij} = B_i - B_j$ and $\Delta D_{ij} = D_i - D_j$. Supplementary Figure S4 shows the physical branch. It computes five features:

$$\phi_{ij}^{\text{phys}} = \log\left(1 + \left[\frac{|\Delta t_{ij}|}{u_{t,ij}},\ z_{t,ij},\ z_{f,ij},\ \frac{|\Delta B_{ij}|}{u_{B,ij}},\ \frac{|\Delta D_{ij}|}{u_{D,ij}}\right]\right), \tag{12}$$

The logarithm acts componentwise. The timing inputs distinguish total normalized delay from delay beyond the receiver-baseline allowance, allowing the scorer to discriminate among admitted pairs.

The semantic branch uses compatibility v_{ij} and concentration r_i:

$$v_{ij} = \log\Big(\max\{K\pi_i^{\mathsf{T}}\pi_j, \epsilon\}\Big), \qquad H(\pi_i) = -\sum_{q=1}^{K} \pi_{iq} \log \pi_{iq}, \tag{13}$$

$$r_i = 1 - \frac{H(\pi_i)}{\log K}, \qquad \phi_{ij}^{\text{sem}} = r_i r_j v_{ij}. \tag{14}$$

We use floor $\epsilon = 10^{-12}$ and convention $0 \log 0 = 0$. Factor K makes uniform evidence neutral. Concentration $r_i \in [0, 1]$ downweights diffuse vectors in Equation (14), but can amplify confidently wrong evidence. The unweighted variant uses v_{ij} directly and is the default linear option based on the initial nominal ablation. New comparisons retain every scorer without test-based selection; entropy weighting remains optional, with no established reliability advantage.

The feature vector is $\phi_{ij} = [\phi_{ij}^{\text{phys}}, \phi_{ij}^{\text{sem}}] \in \mathbb{R}^6$, with each model's stated semantic input. Training means μ and scales σ give $\widetilde{\phi} = (\phi - \mu)/\sigma$; zero-variance components receive scale one. The linear logit is

$$\ell_{ij} = \beta_0 + \beta^{\mathsf{T}} \widetilde{\phi}_{ij}. \tag{15}$$

$$(\beta_0, \beta) = \arg\min_{\beta_0, \beta} \left\{ \sum_{(i,j)\in\mathcal{T}} \left[\log(1 + e^{\ell_{ij}}) - y_{ij}\ell_{ij}\right] + \frac{\|\beta\|_2^2}{2C_{\text{reg}}} \right\}. \tag{16}$$

Training uses all labeled cross-receiver pairs $\mathcal{T}$, $C_{\text{reg}} = 1$, an unpenalized intercept, and no balancing or negative downsampling. Restricted models use fewer inputs. Histogram boosting uses five physical inputs and weighted semantics [48]. All three joint scorers share mapping and decoding rules, with weighted linear results retained as the historical reference.

3.5. Calibration and the interpretation of signed evidence

We freeze the scaler and scorer first. We then fit the temperature on admitted pairs $\mathcal{V}_g$ from independent calibration scenes:

$$T^* = \arg\min_{T\in[0.1,10]} \sum_{(i,j)\in\mathcal{V}_g} [\log(1 + e^{\ell_{ij}/T}) - y_{ij}\ell_{ij}/T]. \tag{17}$$

The calibrated score is $p_{ij}^{\text{cal}} = \text{sigmoid}(\ell_{ij}/T^*)$, where $\text{sigmoid}(x) = 1/(1 + e^{-x})$, and the reported score is $\widehat{p}_{ij} = g_{ij} p_{ij}^{\text{cal}}$. Gaussian-tail true pairs can receive forced zeros, so the gated score is not a calibrated posterior. NLL clips probabilities to $[\epsilon, 1 - \epsilon]$ for finite loss; Brier error uses the reported values.

For decoding, use finite pre-gate signed weights

$$w_{ij} = \operatorname{logit}(p_{ij}^{\mathrm{cal}}) = \ell_{ij}/T^*, \qquad \operatorname{logit}(p) = \log \frac{p}{1-p}. \tag{18}$$

Clipping p^{cal} makes equality to ℓ/T^* an unsaturated rule. Rejected pairs fail feasibility without taking the logit of zero. All positive and negative weights enter the group score.

Proposition 2 (Temperature invariance)**.** *Fix the gates and use unsaturated logits. Break ties deterministically and use a zero merge cutoff. Make pair decisions at* 0.5*. Under these conditions, any common $T > 0$ preserves pair rankings and signs. It also preserves the greedy merge sequence defined below.*

Proof is given in Supplementary Section S10. Clipping, nonzero cutoffs, or different pair thresholds can break invariance. The executable cutoff is 10^{-10}, so nominal before/after partitions are checked separately.

3.6. Constrained event partitions and greedy inference

Operational meaning of consistency.

Membership forms an equivalence relation with receiver exclusivity and complete admission under Equation (10). No score overrides these hard requirements. The feature in Equations (13)–(14) instead supplies signed soft evidence, without identical-label or semantic-threshold requirements.

Joining admitted *a*–*b* and *b*–*c* also asserts *a*–*c*, which must pass. Indirect links cannot repeat a receiver. These checks enforce the contract without certifying event identities or a common source position.

Let $\mathcal{C} = \{C_1, \ldots, C_M\}$ partition all N observations. The cluster count M is unknown. Predicted co-membership is $a_{ij} = \mathbf{1}[i, j \text{ belong to the same } C]$. It must be an equivalence relation and meet the following rules:

$$\sum_{i \in C} \mathbf{1}[s_i = s] \le 1 \quad \forall C \in \mathcal{C},\ s, \qquad a_{ij} = 1 \implies g_{ij} = 1 \quad (s_i \neq s_j). \tag{19}$$

The constraints enforce exclusivity and complete cross-pair admission. We maximize

$$J(\mathcal{C}) = \sum_{C \in \mathcal{C}} \sum_{\substack{i<j \\ i,j \in C}} w_{ij}. \tag{20}$$

Singletons contribute zero. Bernoulli pair factors yield this sum after partition-constant terms are removed, but receiver errors and transitivity couple the factors. Hence J is a composite agreement score, not a joint posterior.

For disjoint current clusters A, B, write $\mathcal{S}(A) = \{s_i : i \in A\}$. Define

$$F(A, B) = \mathbf{1}[\mathcal{S}(A) \cap \mathcal{S}(B) = \varnothing] \prod_{i \in A, j \in B} g_{ij}, \tag{21}$$

$$\Delta(A, B) = \sum_{i \in A, j \in B} w_{ij}, \tag{22}$$

$$\Delta(A \cup B, C) = \Delta(A, C) + \Delta(B, C), \qquad F(A \cup B, C) = F(A, C) \wedge F(B, C). \tag{23}$$

Here F denotes feasibility and $\wedge$ logical AND. Negative cross edges offset positive support in each gain. Figure 4 illustrates an early merge that greedy inference cannot undo.

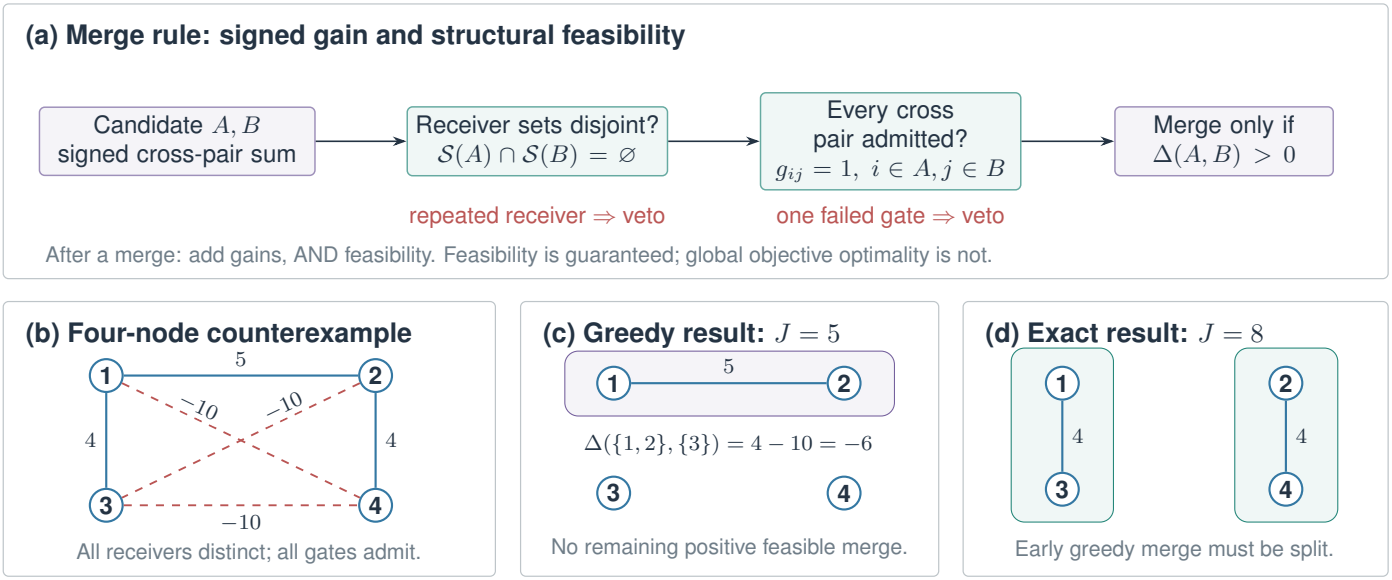


Figure 4. A greedy optimization limit. (a) Positive gains require receiver and all-pair admission checks. (b) Four distinct receivers have all gates admitted, with $w_{12} = 5$, $w_{13} = w_{24} = 4$, and other weights -10. (c) Greedy joins $\{1, 2\}$ and stops at $J = 5$. (d) The optimum $\{1, 3\}, \{2, 4\}$ has $J = 8$. The supplied invariant test verifies this constructed example; it is not an empirical scene.

Algorithm 1 Receiver-exclusive, all-pairs-admissible greedy association

Input: common observations, declared uncertainty, fitted scaler/scorer, T, and k.
Output: a partition of all observations, including singletons.

1. Form cross-receiver features, gate values g_{ij}, and finite pre-gate weights w_{ij}.
2. Initialize singleton clusters and their dense gain and feasibility arrays.
3. Select a feasible pair (A, B) with maximal $\Delta(A, B)$. Stop if no gain exceeds 10^{-10}.
4. Replace A, B by $A \cup B$; update gains and feasibility with Equation (23).
5. Repeat Steps 3–4 and return the remaining clusters.

Proposition 3 (Feasibility, ascent, and termination)**.** *Algorithm 1 returns a receiver-exclusive, all-pairs-admissible partition. Each accepted merge raises J. There are at most $N - 1$ merges. Dense comparisons take $O(N^3)$ work. Storage takes $O(N^2)$.*

Proof is given in Supplementary Section S10. Termination excludes further positive-gain merges but proves neither global optimality nor correct identity. Exact audits maximize $\sum_{i<j} w_{ij} a_{ij}$ over binary a_{ij}, fixing $a_{ij} = 0$ for same-receiver or rejected pairs. For distinct i, j, ν, transitivity requires

$$\begin{aligned} a_{ij} + a_{j\nu} - a_{i\nu} &\le 1, \\ a_{ij} + a_{i\nu} - a_{j\nu} &\le 1, \\ a_{i\nu} + a_{j\nu} - a_{ij} &\le 1. \end{aligned} \tag{24}$$

If i joins j with $a_{ij} = 1$ and j joins ν with $a_{j\nu} = 1$, the first inequality requires $a_{i\nu} = 1$. The MILP has $\binom{N}{2}$ variables and $3\binom{N}{3}$ inequalities. SciPy/HiGHS must prove optimality at requested gap zero; time-limited incumbents are not called exact [49]. A restart heuristic retains greedy decoding and tries up to eight alternative first merges among the strongest positive feasible edges. Each run uses Algorithm 1, retaining the best objective. With L alternatives, work is $O((L + 1)N^3)$ and storage $O(N^2)$, without approximation or accuracy guarantees.

3.7. Completion of partially known class registries

Let $\mathcal{U}$ contain the receivers with missing map entries; receiver 0 retains its known reference map. At receiver s, disclosed anchors form $\mathcal{H}_s$, and $\mathcal{P}(\mathcal{H}_s)$ contains the compatible permutations. Maps remain fixed across adaptation and test. Each comparison freezes its scorer, temperature, and physical gate before map fitting. The objective accepts weighted linear, unweighted linear, or boosted evidence.

Write $\mathcal{C} = (\mathcal{C}_b)_{b\in\mathcal{B}_A}$ for the adaptation partitions. Each $\mathcal{C}_b$ is a whole window partition. A plain C denotes one cluster. The fixed-group objective is

$$J_A(\mathbf{M}, \mathcal{C}) = \sum_{b\in\mathcal{B}_A} \sum_{i<j} a_{ij}(\mathcal{C}_b) w_{ij}(\mathbf{M}), \qquad \mathbf{M} \in \prod_{s\in\mathcal{U}} \mathcal{P}(\mathcal{H}_s). \tag{25}$$

Known receiver maps are held fixed. The fixed-group update is

$$\mathbf{M}^+ \in \arg\max_{\mathbf{M}} J_A(\mathbf{M}, \mathcal{C}). \tag{26}$$

For fixed groups, $J_A = \sum_{s<t} \Psi_{st}(M_s, M_t)$ sums co-grouped report scores by receiver pair. Log compatibility depends on relative permutation, requiring at most 24 evaluations per pair for four classes. Caching scores the full block of up to $24^3 = 13{,}824$ states for three unknown maps; anchors reduce the domain. Supplementary Section S11 derives and verifies this factorization.

Let $U = |\mathcal{U}|$. Let d_s count candidate maps at receiver s. A known map has $d_s = 1$. With a_s correct anchors, an unknown map has $d_s = (K - a_s)!$ candidates. The joint state count is

$$Q = \prod_{s\in\mathcal{U}} d_s \le (K!)^U.$$

With P receiver-pair terms, cached summation costs $O(PQ)$ and stores Q scores. The search remains factorial: four classes with six unknown receivers give 191,102,976 states, requiring 1.42 GiB for float64 scores alone. Six classes with three unknowns give 373,248,000 states. Blocks above one million states are rejected before allocation, while recovery tests use at most 13,824. Larger-block timings provide no larger-vocabulary accuracy evidence.

This solves the conditional block within 10^{-10}, without globally solving maps and events. Cheaper coordinate updates can stall because a source change affects agreement with every other source. Joint changes can recover support unavailable to individual updates, as the strict counterexample in Supplementary Section S11 shows.

Physical-only groups initialize adaptation. Sequential estimation solves Equation (26) once and freezes the map. Alternation re-decodes under current maps, accepts a partition only for strict gain over the previous feasible partition, and then refits maps. Ties retain the old state; at most ten sweeps are allowed. Accepted updates cannot lower J_A, but need not improve reference accuracy. Test maps remain frozen.

Adaptation uses descriptors and provisional groups, never event labels, hidden entries, source positions, or clutter flags. Uniform vectors provide no map information, and incorrect groups can mislead estimation. A known reference resolves the common-renaming ambiguity that compatibility alone cannot remove. Hidden-entry accuracy and event F1 are therefore reported separately.

When an anchored map can be identified.

Use a score graph containing every nonconstant receiver-pair term in the objective. A reference map fixes the names in its connected component. A separate component needs its own known map or enough anchors. Connectivity alone is insufficient: uniform class

vectors provide no correspondence information, and indistinguishable class columns can be exchanged without changing any score.

Let $\psi^*_{st}(R)$ be an ideal fixed-group score for relative permutation R. Let R^*_{st} denote the true relative map. Suppose the score has a positive margin δ_{st} over every wrong relative map. Let the observed score differ from the ideal score by at most ε_{st} for every candidate. Then the conditions

$$\psi^*_{st}(R^*_{st}) - \psi^*_{st}(R) \geq \delta_{st} > 2\varepsilon_{st} \quad \text{for all } R \neq R^*_{st}$$

on every edge of a connected graph ensure a unique anchored optimum (Supplementary Section S15). Ideal scores and error bounds are assumptions, not test-certified quantities. Class coverage, overlapping detections, and correct groups help but do not guarantee recovery or imply that five scenes suffice. The result concerns the mathematical objective; numerical updates use finite precision and cutoff 10^{-10}.

4. Experiments and Results

The controlled benchmark tests the class-probability mechanism, followed by independent mapping comparisons and semantic stress tests. The public check provides auxiliary evidence from measured metadata. Scorer coefficients and gate rules remain fixed throughout these comparisons.

4.1. Controlled descriptor benchmark and evaluation protocol

4.1.1. Synthetic multi-receiver scenes

A nominal scene places four receivers on a randomly rotated circle of radius 2000 m. It contains $E = 14$ hidden *transmission instances*, indexed $e = 0, \ldots, 13$, each with its own time, position, carrier, bandwidth, duration, and anonymous family label. The 14 bursts are not 14 classes, named protocols, reports, or persistent emitters. Independent detection gives zero to four reports per burst; at probability 0.8, the expected genuine total is $14 \times 4 \times 0.8 = 44.8$, before clutter.

Initial positions are drawn in a 3000×3000 m square and start times in a 120 μs window. Carrier and bandwidth are drawn independently before near-collision overrides:

$$\begin{aligned} f_e^{(0)} &= 100\,\text{MHz} + m_e(1.8\,\text{kHz}) + \eta_e, \quad m_e \sim \text{Unif}\{0, \ldots, 5\}, \\ \eta_e &\sim \mathcal{N}(0, (180\,\text{Hz})^2), \qquad\qquad B_e^{(0)} \sim \text{Unif}[20, 60]\,\text{kHz}. \end{aligned} \tag{27}$$

The 1.8 kHz carrier spacing and 20–60 kHz bandwidth deliberately create overlapping descriptors: six centers span only 9 kHz. The 180 Hz event perturbation η_e is shared across reports and cancels from true-pair differences. It is separate from receiver residuals of 250 Hz and report errors of 120–360 Hz (Table 2); the 180 Hz value is not a waveform spectral width.

Initial durations are uniform over 20–80 μs. Each event after the first has probability 0.25 of copying an earlier event's time, carrier, bandwidth, and duration with small perturbations, while retaining independent position and class. These overrides can move parameters beyond the initial ranges.

The collisions create descriptor ambiguity without simulating interference, packet detection, or overlapping-waveform demodulation. All 14 first-scene events and detecting receivers are recorded in `results/event_ledger_r00_test000.csv`.

Detection probability is 0.8 with independent clutter. Clock and oscillator residuals are scene-shared, whereas detector errors are drawn per report. Native frequency units and class permutations are applied before shuffling.

Table 2. Nominal generator and inference settings. Receiver index s starts at zero. Standard deviations describe measurement errors unless marked as shared residual biases. Supplementary Sections S1–S2 give the generation order and stress changes.

Quantity	Configuration
Receivers; transmission instances; classes	$S = 4$; $E = 14$; $K = 4$ anonymous signal-family labels
Nominal carrier-center spacing	1.8 kHz; six centers around 100 MHz, not channel bandwidth
Event carrier perturbation s.d.	180 Hz, shared across reports of one event before collision overrides
Initial event bandwidth; duration	Uniform 20–60 kHz; uniform 20–80 μs, before collision overrides
Detection; clutter	Detection probability 0.8; independent Poisson mean 0.8 per receiver
Timing; carrier measurement s.d.	$(0.6 + 0.4s)$ μs; $(0.12 + 0.08s)$ kHz
Bandwidth; duration measurement s.d.	$(1.5 + 0.3s)$ kHz; $(1 + 0.3s)$ μs
Scene-shared receiver residuals	Clock s.d. 0.8 μs; oscillator s.d. 0.25 kHz; correctly declared nominally
Semantic corruption; missingness	Wrong-class probability 0.12; missing-vector probability 0.10
Reported class confidence	Correct: uniform $[0.60, 0.95]$; incorrect: uniform $[0.45, 0.90]$
Linear scorer; calibration	$C_{\text{reg}} = 1$, maximum 500 iterations; $T \in [0.1, 10]$
Boosted comparator	100 iterations; 15 maximum leaf nodes; learning rate 0.08; $L_2 = 1$
Gate; greedy numerical cutoff	$k = 3.5$; positive merge gain exceeding 10^{-10}

Table 3. Nominal data totals over ten replications. Positive pairs represent a common true event; all listed pairs are cross-receiver.

Split	Scenes	Observations	Pairs	Positive pairs
Training	1200	57,227	1,023,517	63,897
Calibration	400	19,113	342,476	21,395
Test	800	38,451	693,023	43,151

Generated class vectors give the designated family confidence in $[0.60, 0.95]$ when correct and $[0.45, 0.90]$ when wrong, distributing remaining mass equally. Missing vectors are uniform (Table 2). These are synthetic inputs, not trained waveform-classifier outputs; Section 4.13 treats measured records separately.

4.1.2. Scene splits and reproducible implementation

Ten replications independently regenerate the scenes and refit scorers. Replication $r \in \{0, \ldots, 9\}$ uses seed $10^6 + 10^5 r$, split offsets 0, 10,000, and 20,000, and the scene index. Splitting 120/40/80 scenes before pairing prevents event or receiver-residual leakage across splits [47].

Table 3 gives observed counts. The implementation uses NumPy, SciPy, and scikit-learn [48,49]. Original runs used Python 3.13.5, NumPy 2.3.5, SciPy 1.17.0, scikit-learn 1.8.0, and pandas 2.2.3 on Linux with an AMD EPYC 9V74 processor. Added experiments record a Windows environment. Numerical libraries use one thread without a GPU; original timing curves exclude generation, features, fitting, and scoring.

4.1.3. Comparators, ablations, and metrics

We compare timing-only, frequency-only, semantics-only, physical, and six-input linear or boosted scorers. Raw-frequency ablation removes conversion only from the frequency *feature*, retaining the canonical gate and other inputs. Separate variants remove entropy weighting, gating, or temperature scaling.

All decoders share observations and pair evidence. Components join admitted edges above 0.5. Pairwise assignment allows zero-gain dummies and then unions receiver-pair matchings, without optimizing a global multipartite assignment.

Table 4. Event alignment under supplied class maps. All rows use the same frozen joint scorer, physical gate, decoder, and scenes. F1 scores event matches. It does not measure recovery of an unknown vocabulary. Intervals use ten replication-level means.

Input treatment	Event-pair F1 [95% CI]	Receiver conflict	Gate violation
Declared class mapping applied	0.7798 [0.7713, 0.7883]	0.00%	0.00%
Native columns treated as common	0.4521 [0.4366, 0.4676]	0.00%	0.00%
Class evidence set to uniform	0.6438 [0.6290, 0.6587]	0.00%	0.00%

Constrained scorers use Algorithm 1. Ablations remove exclusivity or complete cross-pair admission separately. Without exclusivity, unscored same-receiver edges have zero weight. Stress tests retain nominal fitted parameters.

AUROC and non-interpolated average precision (AP) assess discrimination; Brier error and clipped negative log-likelihood (NLL) assess probabilities [40,45]. Pair tables include rejected pairs, whereas reliability diagrams use admitted pairs in ten equal-width bins.

Partition metrics use *all unordered observation pairs*, so same-receiver false links are penalized. Event-pair F1 is

$$F_1 = \frac{2\,\mathrm{TP}}{2\,\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}, \tag{28}$$

TP, FP, and FN count common, predicted-only, and true-only co-memberships. We also report ARI, B^3 F1, exact visible-event recovery, and clutter attachment [45,46]. Exact recovery requires an event seen at least twice to be recovered without additions or omissions. Receiver violations use non-singleton clusters, while gate violations use predicted within-cluster cross-receiver edges (Supplementary Section S3).

Pair metrics pool 80 scenes per replication; partition metrics first average scene scores. Ten-replication means use 95% intervals $\bar{x} \pm t_{0.975,9} s_x / \sqrt{10}$, with Student t on nine degrees of freedom and replication standard deviation s_x. The F1 estimand is $\frac{1}{10}\sum_{r=1}^{10}\frac{1}{80}\sum_{b=1}^{80} F_{1,rb}$. Joint constrained counts are TP = 30,063, FP = 3589, and FN = 13,088; TP + FN equals the 43,151 test positives (Table 3). Micro F1 is 0.7829 versus scene-macro F1 of 0.7798. Supplementary Section S14 gives all counts; method differences are paired before interval calculation.

The 693,023 dependent pairs are not independent replications. Intervals describe generation and refitting under this model, not variation across field deployments.

4.2. Representation and known-vocabulary alignment checks

The mapping audit reuses 800 scenes and ten frozen joint-linear scorers with fixed gates and decoding. Correctly mapped, bypassed native, and uniform class vectors change the input only, without new scenes or fitting.

Mapped F1 is 0.7798 versus 0.4521 for bypass, a paired difference of 0.3277 with 95% interval $[0.3162, 0.3392]$. Uniform vectors give 0.6438, compared with 0.6813 from separately refitted physical scoring. Input removal and refitting answer different questions. All conditions remain feasible, so structural checks do not correct class meaning.

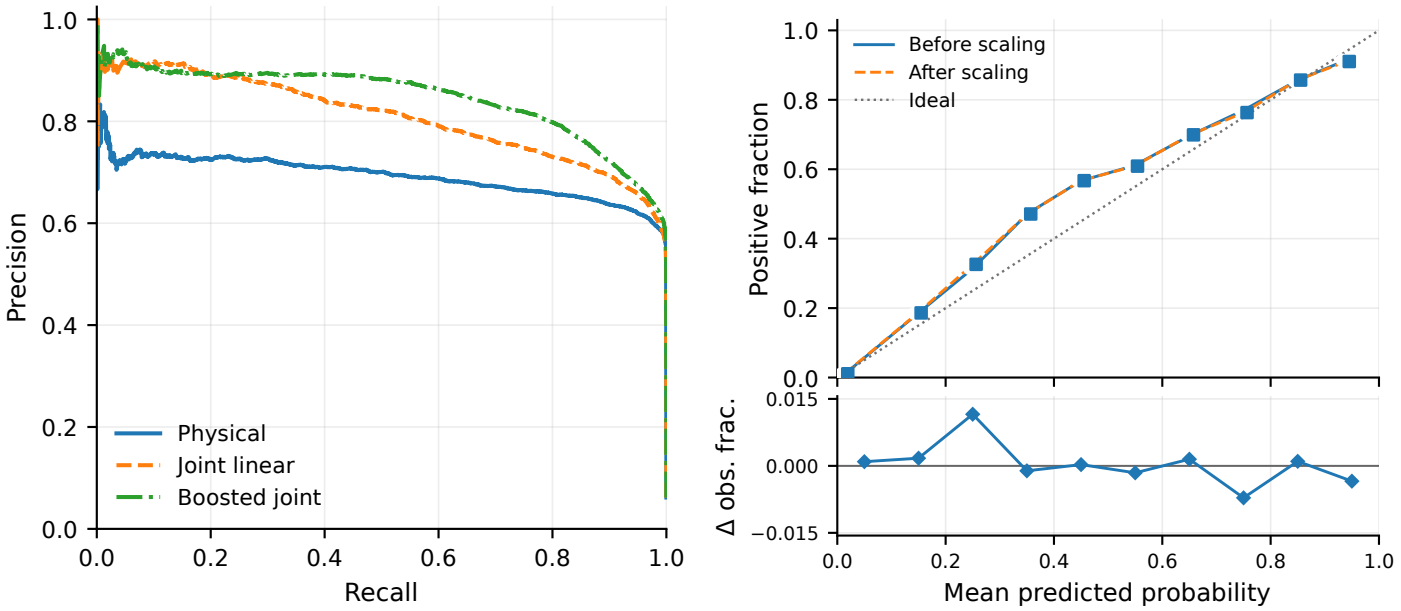
Matched unit and class-order changes preserve all 800 partitions with zero feature or probability difference, checking Equation (5). The 800 re-encodings reuse scenes and do not test independent accuracy or unknown, partial, or hierarchical maps. Outputs are retained in `results/semantic_alignment/`.

4.3. Pair discrimination and confidence quality

Class evidence raises calibrated linear AP from 0.6890 to 0.7979 (Table 5), a paired gain of 0.1089 with 95% interval $[0.1015, 0.1164]$. It resolves some physical ambiguity that

Table 5. Pair results over ten independently fitted replications. Higher AUROC and AP are better. Lower Brier error and NLL are better. T means temperature scaling. AP intervals use variation across replications.

Method	AUROC	AP [95% CI]	Brier	NLL
Timing only	0.9135	0.2864 [0.2799, 0.2928]	0.0469	0.1435
Frequency only	0.8995	0.2550 [0.2476, 0.2624]	0.0502	0.1611
Semantics only	0.7785	0.1607 [0.1583, 0.1630]	0.0563	0.2119
Physical, ungated	0.9831	0.6885 [0.6744, 0.7027]	0.0253	0.0741
Joint linear, ungated	0.9885	0.7963 [0.7878, 0.8048]	0.0211	0.0644
Boosted joint, ungated	0.9910	0.8346 [0.8278, 0.8414]	0.0183	0.0562
Raw-frequency feature	0.9868	0.7757 [0.7665, 0.7850]	0.0227	0.0696
Unweighted semantics, ungated	0.9899	0.8202 [0.8120, 0.8284]	0.0198	0.0611
Joint linear, gated	0.9884	0.7979 [0.7893, 0.8065]	0.0209	0.0638
Joint linear, gated + T	0.9884	0.7979 [0.7893, 0.8065]	0.0208	0.0637
Physical, gated + T	0.9829	0.6890 [0.6747, 0.7032]	0.0250	0.0734
Boosted joint, gated + T	0.9908	0.8343 [0.8274, 0.8412]	0.0183	0.0569



(a) All cross-receiver test pairs.

(b) Admitted pairs of the joint linear model.

Figure 5. Diagnostics for held-out replication 0. (a) Precision–recall curves compare physical, joint linear, and boosted scores with gating and calibration. (b) Ten-bin confidence summaries and their post-minus-pre differences. Horizontal displacement ± 0.006 improves legibility without altering values. Tables provide uncertainty across replications.

moderate single-modality AUROC obscures. Boosting reaches AP 0.8343, indicating useful nonlinear interactions.

Figure 5a compares discrimination; Figure 5b evaluates admitted-pair confidence, matching the temperature-fitting population. Scaling changes paired Brier error by -2.88×10^{-5} with interval $[-5.60 \times 10^{-5}, -1.55 \times 10^{-6}]$: statistically nonzero but numerically modest.

Positive rescaling preserves unsaturated ranking. All 800 executable nominal partitions remain unchanged, consistent with Proposition 2; calibration changes confidence without clustering gain.

Removing concentration weighting raises ungated AP from 0.7963 to 0.8202, showing that diffuse compatibility can remain useful. Raw-frequency AP falls to 0.7757, with other features and the canonical gate partly compensating. Entropy weights are not a universal

Table 6. Scorer–decoder combinations on 800 test scenes. Decoders share each scorer's probabilities and gates; F1 intervals use ten refitted replications. Receiver conflicts are counted per non-singleton cluster and gate violations per predicted within-cluster cross-receiver edge.

Scorer	Decoder	Event-pair F1 [95% CI]	ARI	B^3 F1	Exact	Rx conflict	Gate violation
Physical linear	Components	0.6594 [0.6472, 0.6717]	0.6369	0.8432	0.4428	22.2%	0.5%
Physical linear	Assignment + union	0.7145 [0.7038, 0.7252]	0.6981	0.8543	0.4981	12.7%	0.2%
Physical linear	Constrained	0.6813 [0.6658, 0.6968]	0.6683	0.8129	0.4072	0.0%	0.0%
Joint linear	Components	0.6764 [0.6641, 0.6887]	0.6549	0.8540	0.4909	21.1%	0.7%
Joint linear	Assignment + union	0.7797 [0.7741, 0.7853]	0.7671	0.8898	0.5973	10.5%	0.1%
Joint linear	Constrained	0.7798 [0.7713, 0.7883]	0.7704	0.8677	0.5375	0.0%	0.0%
Unweighted joint	Components	0.6889 [0.6780, 0.6998]	0.6684	0.8597	0.5067	19.7%	0.6%
Unweighted joint	Assignment + union	0.8025 [0.7974, 0.8075]	0.7915	0.8993	0.6238	8.6%	0.1%
Unweighted joint	Constrained	0.7983 [0.7892, 0.8074]	0.7897	0.8784	0.5652	0.0%	0.0%
Boosted joint	Components	0.7016 [0.6888, 0.7145]	0.6819	0.8676	0.5354	18.6%	0.9%
Boosted joint	Assignment + union	0.8130 [0.8063, 0.8196]	0.8024	0.9072	0.6506	8.6%	0.2%
Boosted joint	Constrained	0.8238 [0.8181, 0.8295]	0.8160	0.8928	0.6189	0.0%	0.0%

reliability estimate, and this frequency ablation does not remove the full observation contract.

4.4. Event partitions and the role of explicit constraints

Joint constrained F1 is 0.7798 versus physical-only F1 of 0.6813 (Table 6; Figure 6), a paired gain of 0.0985 with interval $[0.0887, 0.1083]$. Components yield F1 0.6764, recall 0.8985, and precision 0.5703 because unchecked chains can create oversized or receiver-conflicting groups.

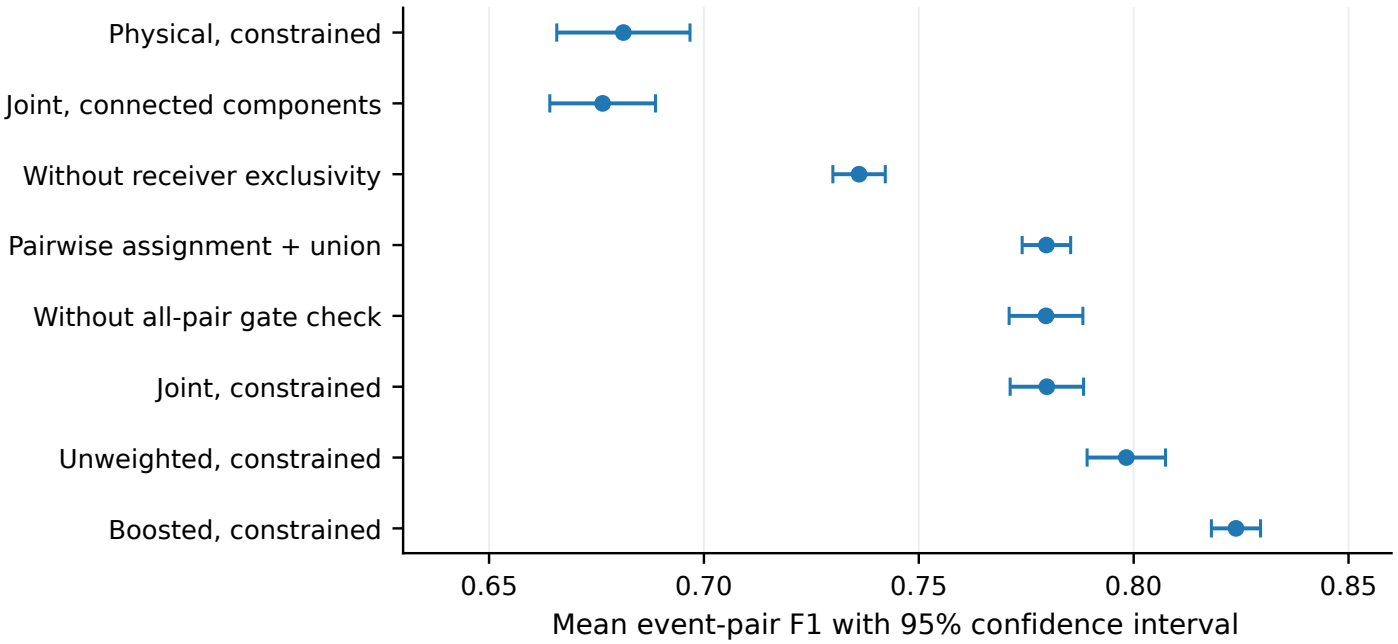


Figure 6. Event-pair F1 for each scorer and decoder. Points show means of ten replication-level averages. Bars show 95% t intervals. All methods use the same held-out scenes. Unweighted semantics and boosted scoring both give higher F1 than the joint constrained reference.

Receiver conflicts affect 21.14% of non-singleton component clusters and 10.49% of assignment clusters. Algorithm 1 prevents receiver and gate violations, but its precision 0.8932 and recall 0.6972 leave many true pairs unjoined.

Table 7. Gate-width test on 200 nominal scenes: ten replications times the first 20 test scenes. Retained pair fraction counts admitted cross-receiver pairs. True-pair recall uses only pairs from a common true event.

k	True-pair recall	Retained pair fraction	Event-pair F1 [95% CI]
1	0.6837	0.0885	0.5975 [0.5806, 0.6145]
2	0.9585	0.1354	0.7569 [0.7394, 0.7744]
3	0.9979	0.1621	0.7808 [0.7620, 0.7996]
3.5	0.9997	0.1772	0.7822 [0.7635, 0.8009]
5	1.0000	0.2254	0.7825 [0.7640, 0.8010]

Assignment gives similar F1 (0.7797) and higher exact recovery (0.5973 versus 0.5375), while producing some infeasible groups. Accuracy and structural validity must therefore be assessed together.

Table 6 crosses four scorers and three decoders. Boosted F1 is 0.7016 for components, 0.8130 for assignment, and 0.8238 for constraints. The constrained-minus-assignment gain is 0.0108 with 95% interval $[0.0020, 0.0196]$, while assignment retains 8.57% receiver conflicts.

Physical-only constrained F1 is 0.6813 versus assignment F1 of 0.7145, a difference of -0.0332 with interval $[-0.0439, -0.0225]$. Joint and unweighted intervals include zero. Constraint accuracy depends on evidence, while feasibility is common to all constrained scorers. The 9600 outcomes and frozen probabilities reproduce the retained results.

Removing exclusivity lowers F1 to 0.7361 and repeats receivers. Removing complete admission gives F1 0.7796 but permits rejected within-group edges, making this check principally a nominal feasibility safeguard.

Unweighted and boosted scoring give F1 0.7983 and 0.8238. The former motivates the simpler linear default; weighted scoring remains the historical control. Independent validation retains all three semantic scorers for event and missing-map tests.

4.5. Admission sensitivity and distribution shifts

4.5.1. Gate width and irreversible pair removal

The frozen-scorer sweep uses the first 20 scenes per replication (Table 7). At $k = 1$, true-pair recall is 68.37% and F1 is 0.5975; later semantics cannot restore rejected edges. At $k = 3.5$, recall is 0.9997 with 17.72% of candidates retained. Increasing k to 5 adds competition with little gain. At $k = 3.5$, F1 0.7822 differs from 0.7798 because the sweep uses 20 rather than 80 scenes.

4.5.2. Incorrect semantics and misspecified receiver uncertainty

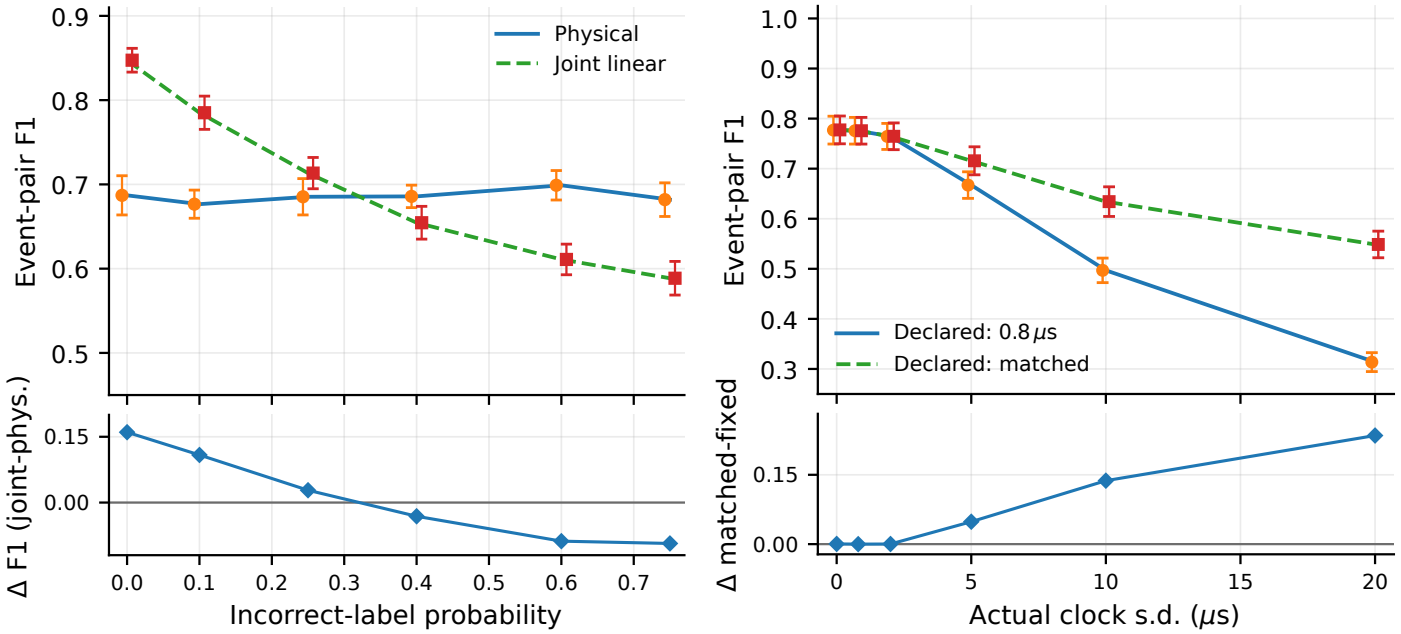
The frozen-model grid has 34 settings: 12 named conditions, six semantic-error probabilities, 12 actual-versus-declared clock combinations, and four scene sizes. Each uses 20 scenes per replication with seeds disjoint from nominal splits. Reusing stress schedules couples comparable draws; the 34 settings are controlled interventions, not separate measured datasets.

The stress reference gives F1 0.7758 (Table 8, first row). At semantic-error probability 0.45, joint F1 is 0.6308 versus physical F1 of 0.6727 (Table 8). One confidently wrong receiver reduces joint F1 to 0.5117: its family is shifted to $(q_e + 1) \bmod K$ with confidence 0.99, remaining mass 0.01, and missingness disabled. Equation (14) cannot detect these concentrated errors, defining a failure regime in Figure 7a.

Actual oscillator s.d. 1.5 kHz versus declared 0.25 kHz reduces gate recall to 0.5606 and F1 to 0.4110. With clock s.d. 20 μs, declaring 0.8 μs gives F1 0.3136; matching the declaration raises it to 0.5486.

Table 8. Named stress tests with 200 scenes per condition. The separated first row is the stress-seed reference. It differs from the nominal test subset in Table 7. The semantic-error row uses error probability 0.45. All other settings follow the stated intervention. The scorer, scaler, gate multiplier, and temperature stay fixed. F1 is event-pair F1. The last column is true-pair gate recall.

Test condition	Physical event F1	Joint event F1 [95% CI]	Gate recall
Reference condition on the stress seeds			
Nominal stress reference	0.6692	0.7758 [0.7491, 0.8024]	0.9996
Interventions on the same seed schedule			
28 events	0.6754	0.7601 [0.7478, 0.7724]	0.9997
Detection probability 0.55	0.6770	0.7652 [0.7472, 0.7831]	0.9993
Clutter mean 2.5 / receiver	0.6881	0.7862 [0.7754, 0.7970]	0.9997
Semantic error probability 0.45	0.6727	0.6308 [0.6093, 0.6523]	0.9992
Missing semantics probability 0.65	0.6692	0.6612 [0.6369, 0.6855]	0.9996
Clock s.d. 6 μs; declared 0.8 μs	0.5303	0.6255 [0.5996, 0.6513]	0.9235
Frequency bias s.d. 1.5 kHz; declared 0.25 kHz	0.3265	0.4110 [0.3845, 0.4375]	0.5606
Additional positive timing delays	0.6377	0.7145 [0.6960, 0.7330]	0.9549
Variance-matched t_3 timing noise	0.6885	0.7786 [0.7632, 0.7941]	0.9993
Fifth receiver, not used in training	0.6743	0.7780 [0.7626, 0.7934]	0.9998
One confidently wrong receiver	0.6692	0.5117 [0.4970, 0.5264]	0.9996



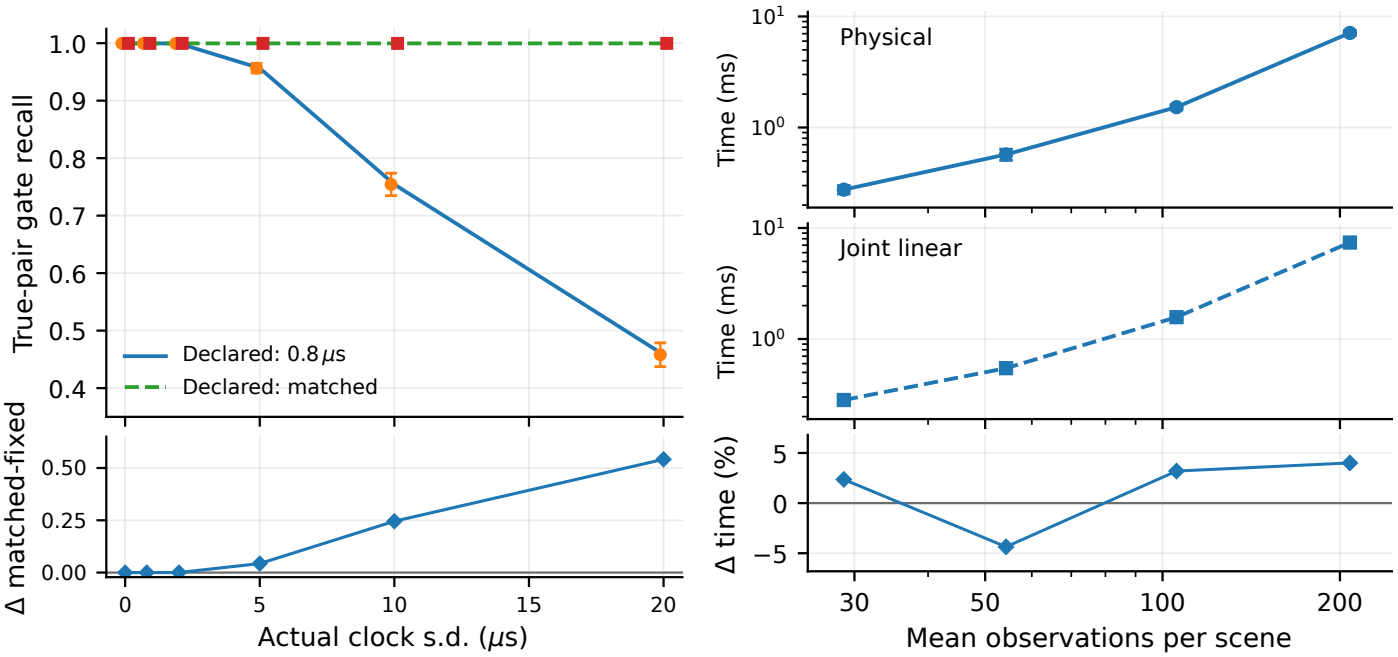
(a) Increasing semantic-label error.

(b) Increasing receiver clock residual.

Figure 7. Frozen-model stress tests. Upper panels show ten-replication means and 95% intervals; lower panels enlarge method differences on separate scales. Symbols shift by ± 0.007 in (a) and $\pm 0.12\,\mu$s in (b), with trend lines at true values. Incorrect class confidence can harm joint scoring; correct clock declarations reduce but do not remove timing ambiguity.

Gate recall rises from 0.4581 to 0.9996 (Figure 8a), but broader uncertainty also admits more competing matches. Restoring coverage alone does not restore accurate grouping.

Positive delays reduce gate coverage, while variance-matched heavy tails have a smaller effect here. The fifth-receiver test changes network size within the generator, not arbitrary receiver domains. Clutter and detection changes alter visibility and metric denominators, so higher scores do not establish a benefit from clutter or missed detections.



(a) Gate coverage under clock misspecification.

(b) Measured dense-decoder runtime.

Figure 8. Coverage and runtime diagnostics. (a) Fixed and matched uncertainty, with their recall difference below; symbols shift by $\pm 0.12\,\mu$s only for legibility. (b) Physical and joint runtimes use equally scaled log axes and a percentage-difference strip. Event counts 8, 16, 32, and 64 yield different report counts. Timings exclude generation and scoring; ten-replication intervals are specific to this hardware.

4.6. Computational cost, exact audit, and reproducibility

Figure 8b shows increasing dense-decoder work. Four sizes do not establish an asymptotic exponent; Proposition 3 provides the $O(N^3)$ bound. The implementation retains dense matrices, so no sparse-implementation speedup is demonstrated.

Across 60 scenes with 11–19 reports, greedy and exact objectives agree in 47 cases. Relative gap $(J^* - J_g)/\max(|J^*|, 10^{-9})$ averages 1.71% and peaks at 16.82%. Mean F1 is 0.7053 for greedy and 0.7182 for exact inference.

This comparison holds the *same learned objective* fixed. It measures small-instance optimization loss, not larger-instance near-optimality or a monotone relation to true F1. Figure 4 illustrates the greedy limit; detailed gaps and selective-pair risk–coverage curves remain in the supplement.

Audits regenerate all 2400 nominal scenes, verify splits and counts, recompute summaries and invariants, and compare predictions with fitted parameters. Stored outputs retain 6400 original rows and the separate 9600-row factorial, whose accuracy and feasibility are recomputed from saved labels.

Scientific outputs repeat within floating-point tolerance, while runtimes depend on execution conditions. Subsequent assumption tests remain descriptor-based.

4.7. Single-receiver exploratory mapping probe

Table 9 reports an exploratory test that hides three entries at one receiver. One existing calibration scene recovers 26/30 entries across ten repetitions; five or 40 scenes recover 30/30. Because these descriptors were previously used for temperature fitting, they do not form an independent adaptation set. Supplementary Section S9 gives the protocol.

The measured-input probe hides an artificial swap of decoded SF7/SF12 meanings at one LoED gateway. Adaptation uses 3 August. Testing uses 2 September. Recovering the two entries restores reference-packet F1 from 0.1270 to 0.9961. This is code registration in

Table 9. Optional missing-map test with the original scores and decoder. Controlled results average ten repetitions. Public results use one later test capture. Mapping accuracy counts only hidden entries. Controlled rows use event-pair F1. Public rows use reference-packet F1.

Probe / adaptation	Method	Map (%)	Pair F1
Controlled / 1 scene	Unresolved completion	33.3	0.7026
Controlled / 1 scene	Sequential map fit	86.7	0.7639
Controlled / 1 scene	Map–event feedback	86.7	0.7639
Controlled / 40 scenes	Sequential map fit	100.0	0.7798
Controlled / 40 scenes	Map–event feedback	100.0	0.7798
Controlled / 40 scenes	Complete-map reference	100.0	0.7798
Public / 1 capture	Unresolved completion	0.0	0.1270
Public / 1 capture	Sequential map fit	100.0	0.9961
Public / 1 capture	Map–event feedback	100.0	0.9961
Public / 1 capture	Complete-map reference	100.0	0.9961

The controlled test hides three entries at one receiver. The public test hides only SF7/SF12 at one gateway. Five-scene controlled adaptation also reaches 100% and F1 0.7798. The public test has 8423 reports. The original public study pools 23,677 reports. Feedback does not improve on sequential fitting.

decoded metadata. The experiment contains no measured heterogeneous class posteriors. Sequential and alternating estimates tie. The independent multi-receiver experiment below provides the stronger controlled mapping test.

4.8. Map robustness and multi-receiver completion

4.8.1. Incorrect relative mappings and shared renaming

The 800-scene wrong-map audit freezes the joint scorers and changes 0, 2, 4, 8, 12, or 16 of 16 entries through bijections. Three interventions per budget are averaged within replication. At 25% corruption, F1 falls from 0.7798 to 0.6422; at 50%, it is 0.5474 (Figure 9a). All groups remain feasible, showing that constraints cannot repair class meaning.

The curve is nonmonotone because random permutations can retain relative agreement. A shared renaming changes every absolute entry but preserves all 800 partitions. Event F1 therefore cannot establish absolute naming accuracy.

4.8.2. Independent adaptation and conditional map optimization

Each replication supplies 40 adaptation scenes disjoint from all original splits and stress seeds, without refitting. One to three maps are hidden while receiver 0 stays known; unknown receivers receive zero to two anchors. Fixed prefix budgets 1, 5, and 40 are all reported, without test-selected stopping. The 27 configurations use ten seeds and reuse the initial 800 tests. This analysis followed inspection of those tests; the subsequent comparison uses independent tests.

For three unknown maps without anchors, one scene recovers 75/120 entries with F1 0.6787. Five scenes recover 120/120 and restore F1 0.7798 (Table 10), as do forty. Budgets two to four were untested, so five is only the first successful tested value. The ten map interventions do not constitute 120 independent Bernoulli trials or establish larger-vocabulary sample complexity.

Five-scene coordinate fitting recovers 60% of entries with F1 0.7171, whereas exact fitting recovers all. In replication 0 with fixed groups, every single-map gain is nonpositive, but joint updating raises the objective from 201.0133 to 235.4834. Coupling between maps causes this trap despite complete source-domain enumeration. Exact sequential and alternating fitting tie in every tested configuration, with no added event-update gain.

Table 10. Independent adaptation without target anchors. The reference map remains known. Counts sum ten interventions; F1 uses the same 800 tests. Coordinate updates one map at a time; sequential and alternating methods solve the exact conditional block. All receiver and gate violations are zero.

Adaptation scenes	Unknown sources	Correct hidden entries	Sequential F1	Feedback F1 [95% CI]	Coordinate F1	Rx / gate
1	1	35/40	0.7604	0.7604 [0.7280, 0.7929]	0.7604	0 / 0
1	2	61/80	0.7163	0.7163 [0.6628, 0.7697]	0.6941	0 / 0
1	3	75/120	0.6787	0.6787 [0.6272, 0.7301]	0.6279	0 / 0
5	1	40/40	0.7798	0.7798 [0.7713, 0.7883]	0.7798	0 / 0
5	2	80/80	0.7798	0.7798 [0.7713, 0.7883]	0.7798	0 / 0
5	3	120/120	0.7798	0.7798 [0.7713, 0.7883]	0.7171	0 / 0
40	1	40/40	0.7798	0.7798 [0.7713, 0.7883]	0.7798	0 / 0
40	2	80/80	0.7798	0.7798 [0.7713, 0.7883]	0.7798	0 / 0
40	3	120/120	0.7798	0.7798 [0.7713, 0.7883]	0.7133	0 / 0

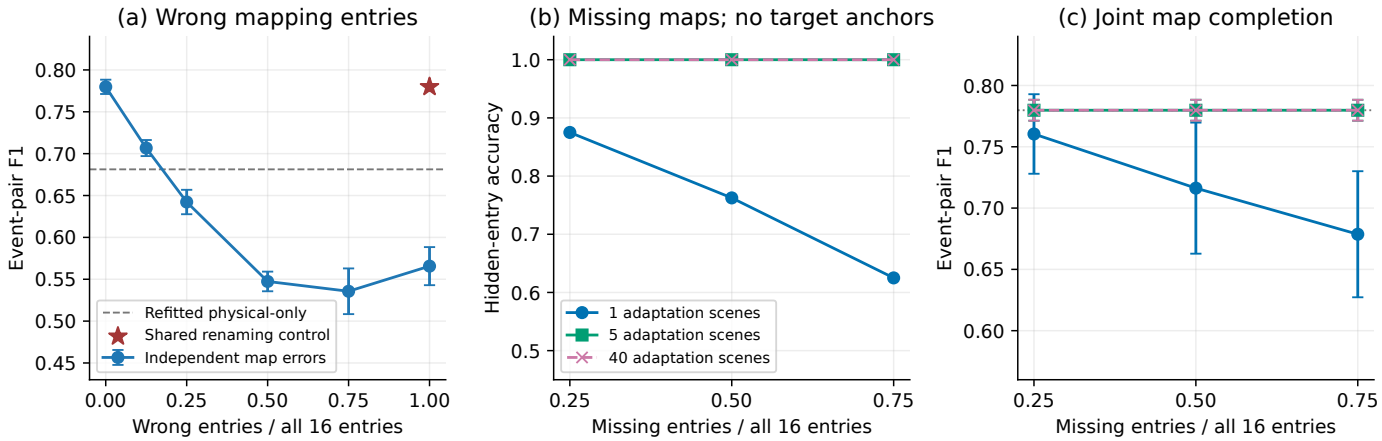


Figure 9. Mapping interventions. (a) Relative-map errors harm association, whereas shared renaming preserves it. (b,c) Missing fractions use all 16 entries, with the reference known and no target anchors. Five- and 40-scene curves coincide. F1 bars use replication means; hidden-entry fractions are descriptive. All violation rates are zero.

Schema tests reject many-to-one maps, three-column coarse vectors, and a fifth unknown class with positive mass. This defines the implementation boundary, not open-set or hierarchical recovery. Supplementary Section S11 records anchors and maps.

4.9. Independent comparison of missing-map estimators

Independent validation uses 400 adaptation and 800 test scenes, with ten replications of 40 and 80 scenes. Prefix budgets are 1, 5, and 40. Receiver 0 remains known; three unanchored four-class maps are hidden. All solvers share physical-only groups and frozen parameters. Reconstructing the boosted model from original training with its saved temperature reproduces all 800 partitions, with maximum probability difference below 4.5×10^{-11}.

Reference-only fitting uses receiver 0, while coordinates update one map at a time. Adapted spectral synchronization [11] selects pairwise relative maps, then uses a leading eigenspace and Hungarian rounding. Spectral-plus-coordinate refines the full RF objective. Exact fitting retains all pair-table scores; sequential fitting solves once, whereas alternation also proposes groups. All maps freeze before testing (Supplementary Section S15).

Five adaptation scenes give exact recovery of all 120 hidden entries for every scorer. Weighted, unweighted, and boosted F1 are 0.7786, 0.7997, and 0.8232, matching known-map references. These 120 entries belong to ten interventions, not 120 independent trials (Table 11).

Table 11. Missing-map estimators using five adaptation and 80 test scenes per replication. Three four-class maps are unknown; only the reference is known. Counts sum ten interventions; F1 intervals use ten replication means. Sequential exact, alternating exact, and known-map results coincide. All outputs satisfy receiver and gate constraints.

Scorer	Map estimator	Correct / hidden	Event-pair F1 [95% CI]
Weighted linear	Unresolved	21/120	0.5316 [0.5119, 0.5514]
Weighted linear	Reference only	113/120	0.7565 [0.7224, 0.7906]
Weighted linear	Spectral sync.	114/120	0.7712 [0.7548, 0.7877]
Weighted linear	Spectral + coordinate	114/120	0.7712 [0.7548, 0.7877]
Weighted linear	Coordinate	70/120	0.7005 [0.6541, 0.7469]
Weighted linear	Exact block	120/120	0.7786 [0.7718, 0.7853]
Unweighted linear	Unresolved	21/120	0.4695 [0.4452, 0.4937]
Unweighted linear	Reference only	118/120	0.7912 [0.7713, 0.8110]
Unweighted linear	Spectral sync.	120/120	0.7997 [0.7913, 0.8080]
Unweighted linear	Spectral + coordinate	120/120	0.7997 [0.7913, 0.8080]
Unweighted linear	Coordinate	66/120	0.7078 [0.6478, 0.7678]
Unweighted linear	Exact block	120/120	0.7997 [0.7913, 0.8080]
Boosted	Unresolved	21/120	0.4458 [0.4230, 0.4686]
Boosted	Reference only	114/120	0.7995 [0.7645, 0.8345]
Boosted	Spectral sync.	114/120	0.8134 [0.7923, 0.8344]
Boosted	Spectral + coordinate	114/120	0.8134 [0.7923, 0.8344]
Boosted	Coordinate	69/120	0.7290 [0.6754, 0.7826]
Boosted	Exact block	120/120	0.8232 [0.8183, 0.8282]

Exact-minus-spectral F1 is 0.0073 for weighted scoring with interval $[-0.0093, 0.0239]$, and 0.0099 for boosting with $[-0.0125, 0.0322]$. Unweighted results coincide, so no general superiority is established. At 40 scenes, reference-only, spectral, and exact fitting recover all maps, while coordinates can stall. Sequential and alternating maps coincide in all 90 scorer–budget–replication settings.

Two same-class conditions hide maps using five adaptation and 40 test scenes per replication. Exact fitting recovers every entry, yet at copy probability 0.65 the scorers give F1 0.4450, 0.4504, and 0.4537, equal to known-map references. Correct registration cannot distinguish nearby same-family events; the next test supplies all maps to isolate this ambiguity.

A five-scene weighted case has 160 grouped pairs and 13,824 states. Cached enumeration takes 2.44 ms versus 1413.08 ms for full reevaluation, with identical optima in three-run medians. A six-class, two-unknown kernel has 518,400 states, taking 377.79 ms for tables and 1.76 ms for search. These timings do not establish larger-vocabulary recovery; Supplementary Section S15 gives memory and allocation limits.

4.10. Class evidence under harder descriptor conditions

Eight conditions use 400 new scenes each, with ten replications of 40 scenes. Four frozen scorers receive correct maps to isolate class-evidence quality. Semantic interventions preserve physical reports; dense conditions share a separate matched physical reference. Supplementary Section S15 specifies probability models, seeds, and intervals.

Nominal near-collisions have independent classes, with agreement probability 1/4: 312 of 1303 copied-event parent pairs match in the new sample. Forcing copies to share the parent's class lowers weighted, unweighted, and boosted F1 from 0.7747, 0.7925, and 0.8226 to 0.7010, 0.7096, and 0.7279, respectively. Physical-only F1 remains 0.6773, attributing part of the semantic gain to class independence.

Increasing copy probability from 0.25 to 0.65 gives boosted F1 of 0.6335 with independent classes and 0.4482 with shared classes, versus physical-only F1 of 0.4460. The paired

Table 12. Semantic stress tests: mean event-pair F1 across ten replication means, with 400 scenes per row and known maps. Semantic interventions preserve physical observations; dense rows share a separate physical reference. Error marginals match in law, though realized counts can differ. All outputs satisfy receiver and gate constraints. Supplementary Section S15 gives full intervals.

Condition	Physical	Weighted linear	Unweighted linear	Boosted
Nominal independent classes	0.6773	0.7747	0.7925	0.8226
Same-class near collisions	0.6773	0.7010	0.7096	0.7279
Dense near collisions / independent classes	0.4460	0.5831	0.6210	0.6335
Dense near collisions / same class	0.4460	0.4408	0.4471	0.4482
Class prior 0.7/0.1/0.1/0.1	0.6773	0.7451	0.7606	0.7852
Receiver-specific confusion and confidence	0.6773	0.7601	0.7822	0.7992
Independent errors / matched marginals	0.6773	0.7714	0.7905	0.8184
Event-shared errors / matched marginals	0.6773	0.8334	0.8826	0.9081

boosted-minus-physical gain is 0.0021 with interval $[-0.0113, 0.0155]$; both linear intervals also include zero. Correct registration offers little event discrimination here.

Class imbalance and receiver-specific classifiers reduce gains (Table 12). The latter retain error probability 0.12 but vary confusion destinations and confidence ranges, separating correct column registration from reliable recognition.

Sharing one error flag and wrong family across an event preserves marginal laws from an independent-error control. Weighted F1 rises from 0.7714 to 0.8334, a paired gain of 0.0620 with interval $[0.0532, 0.0708]$. Agreement on a wrong family can preserve event compatibility, so event F1 does not certify class correctness or establish a universal effect of correlated errors.

4.11. Risk under departures from the observation assumptions

The first 20 tests per replication support nine interventions, two frozen scorers, and three decoders, totaling 10,800 outcomes. Clock equicorrelations 0.8 and -0.25 compare diagonal and matched declarations. Their small effects here do not establish robustness to arbitrary dependence.

Adding exponential delay with mean 8 μs to 30% of genuine reports lowers gate recall from 0.9997 to 0.9353 and constrained F1 from 0.7822 to 0.7122. This is descriptor contamination, not measured NLOS validation.

The alarm test adds 12 false reports by copying three templates to all four receivers. Templates share time, carrier, bandwidth, duration, and canonical class vectors before independent Gaussian timing jitter of s.d. 0.2 μs and frequency jitter of 120 Hz. Native classes are re-encoded. The independent control uses separate templates with matched counts and marginal pool. Each false report has singleton truth; no common interfering waveform is generated.

Correlated false reports have clutter attachment 0.7821 versus 0.7061 in the independent control, yet F1 is also higher at 0.6023 versus 0.5105 (Table 13). Both produce feasible false groups; dependence does not uniformly reduce F1.

With probability 0.2, a genuine report receives a same-receiver copy carrying its transmission label. The original stays unchanged; the copy gets independent Gaussian timing jitter of s.d. 0.15 μs and frequency jitter of 40 Hz, while bandwidth, duration, and class probabilities remain copied. These are detection records, not simulated waveform segments. Constrained F1 is 0.6009.

Let n_{es} count reports of event e at receiver s. The scene-level pair-recall ceiling under receiver exclusivity alone is

$$R_{\max} = \frac{\sum_e \sum_{s<t} \min(n_{es}, n_{et})}{\sum_e \binom{\sum_s n_{es}}{2}}.$$

Table 13. Observation-assumption tests with ten replications of 20 scenes. Rx/gate entries are violation percentages; constrained zeros follow by construction. Clutter attachment counts false reports in non-singletons. The recall ceiling ignores gates and scores.

Intervention	Event-pair F1 [95% CI]	Gate recall	Clutter attachment	Recall ceiling	Rx / gate
Nominal subset	0.7822 [0.7635, 0.8009]	0.9997	0.0067	1.0000	0 / 0
Positive corr.; diagonal declaration	0.7816 [0.7618, 0.8014]	0.9997	0.0067	1.0000	0 / 0
Positive corr.; matched covariance	0.7823 [0.7627, 0.8019]	0.9997	0.0067	1.0000	0 / 0
Negative corr.; diagonal declaration	0.7796 [0.7612, 0.7979]	0.9997	0.0057	1.0000	0 / 0
Negative corr.; matched covariance	0.7798 [0.7609, 0.7986]	0.9997	0.0057	1.0000	0 / 0
Extra positive delay	0.7122 [0.6946, 0.7298]	0.9353	0.0056	1.0000	0 / 0
Independent false alarms; same count	0.5105 [0.4993, 0.5217]	0.9997	0.7061	1.0000	0 / 0
Correlated false alarms	0.6023 [0.5868, 0.6178]	0.9997	0.7821	1.0000	0 / 0
Fragmented reports	0.6009 [0.5857, 0.6161]	0.9998	0.0084	0.6529	0 / 0

The denominator includes same-receiver fragment pairs, which exclusive groups cannot recover. Each receiver pair permits at most $\min(n_{es}, n_{et})$ links, attainable by layered grouping without gates. Empty-positive scenes receive ceiling zero. Figure 10b averages scene ceilings within replication and then across ten replications. The fragment value 0.6529 bounds recall, not F1 (Supplementary Section S12).

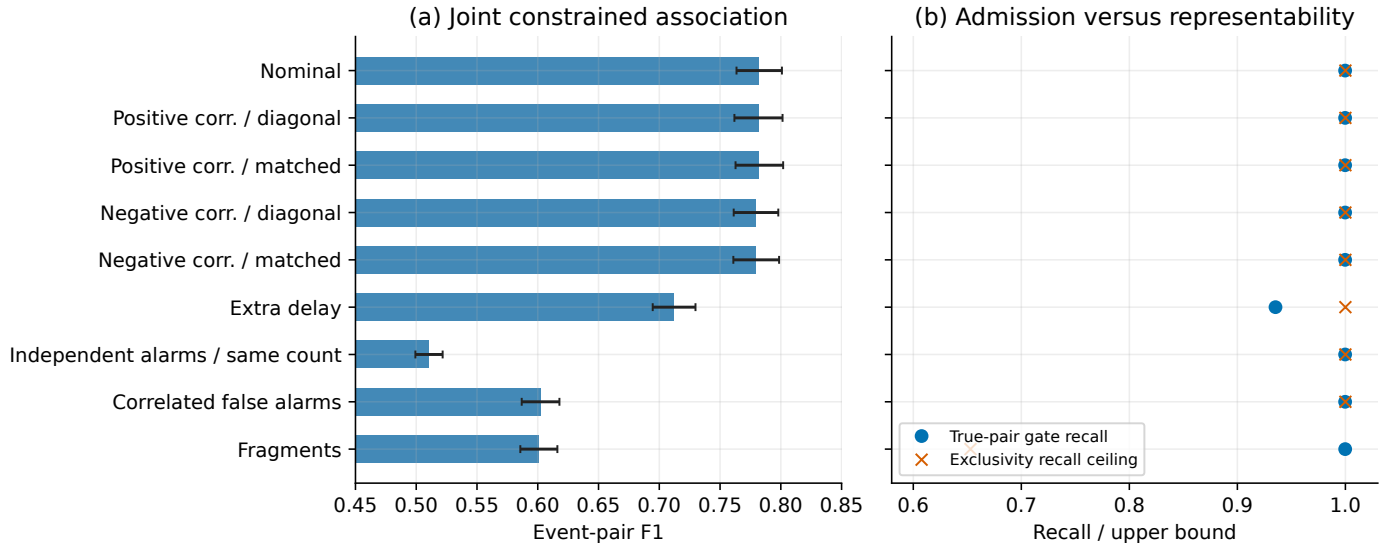


Figure 10. Assumption failures under frozen evidence. (a) Event F1 with 95% replication intervals. (b) Gate recall and the exclusivity-only recall ceiling. Fragments can be unrepresentable despite admission, and correlated false reports can be feasible. Alarm controls have equal added-report counts.

4.12. Medium-size optimization audit

Report count alone selects the first two tests with at least 40 reports per replication. Nested random subsets of 20, 30, and 40 yield 60 MILPs under the frozen greedy objective. All 60 prove optimality within five-second limits at requested gap zero. Records retain incumbents, bounds, status, nodes, feasibility, and construction-plus-solve time.

Greedy mean objective losses are 0.18%, 0.21%, and 0.46%, without implying a general bound. At 40 reports, restarts reduce loss to 0.11%; greedy, restart, and exact F1 are 0.7974, 0.8098, and 0.8142 (Table 14). Better objective search helps this sample but does not establish event truth.

Table 14. Medium-size audit against proven optima. Times are medians in the review environment, including MILP construction. Every decoder is feasible; unproven incumbents are not labeled exact.

N	Proven optima	Greedy gap	Restart gap	Greedy F1	Restart F1	MILP F1	Median ms: greedy / restart / MILP
20	20/20	0.2%	0.0%	0.7689	0.7750	0.7750	0.11 / 0.53 / 5.07
30	20/20	0.2%	0.0%	0.7621	0.7649	0.7649	0.14 / 0.94 / 14.63
40	20/20	0.5%	0.1%	0.7974	0.8098	0.8142	0.22 / 1.52 / 35.32

Table 15. LoED sample and inclusion counts. A visible reference packet has reports at two or more gateways. Eligible test reports come from two gateways. The original dataset covers nine gateways. This experiment does not use all nine.

Split	Files	Raw reports	Eligible reports	Visible reference packets
Training	3	260,231	39,447	257
Calibration	1	25,659	17,252	434
Test	2	40,980	23,677	1,417

Fixed-registry processing time.

Fresh report-path timing includes feature construction, scoring, and decoding for ten scenes at each of 40, 80, and 120 reports. Greedy medians are 0.64, 1.80, and 3.63 ms; triangle MILP medians are 34.56, 284.37, and 956.65 ms. All 30 solves prove optimality. The one-second cap excludes construction, which raises the maximum path to 1014.16 ms. These implementation timings assume no deployment deadline and exclude waveforms and offline map fitting. At 40 reports, greedy F1 is 0.8068 versus MILP F1 of 0.7781. Supplementary Section S15 records timings, solver status, and memory scope.

4.13. Auxiliary public-recording metadata check

4.13.1. Source, inclusion rules, and independent capture blocks

We use the publisher-selected six-file urban LoRaWAN sample [16,17], released under CC BY 4.0 with recorded publisher MD5 and download SHA-256. Training uses 3 March 2019, 2 April 2019, and 2 May 2020; calibration uses 3 July 2020; testing uses 3 August and 2 September 2020. Recorded timestamps, including those preceding filename dates, are retained.

SHA-256 of the complete decoded payload defines the operational packet reference for labels and supervised fitting. Eligible records are CRC-valid unconfirmed uplinks with valid base64; message type comes from the payload header, avoiding the CSV's decimal-rendered binary labels. After exact deduplication, we exclude groups with repeated gateways, spans above 5 s, or identities shared across splits.

These rules remove obvious repetitions without independently proving one transmission. We therefore evaluate *reference-packet association*, with split totals in Table 15 and per-file exclusions in Supplementary Section S7.

Repeated payloads at individual gateways explain many 2019 exclusions. Results therefore concern identifiable receiver-exclusive references, not all traffic; CRC-invalid reports have no independent clutter labels. Reference hashes are stored separately, and scoring excludes addresses, frame counters, payloads and hashes, ports, and row indices.

4.13.2. Measured-input adaptation and frozen parameters

LoED has no waveform-derived uncertainty or class posteriors, so the reduced model uses reception time t_i, channel F_i, and decoded spreading factor L_i. Clock corrections $\widehat{b}_s$ use matching training references.

Gateway pairs need ten references for clock fitting. Median differences define a weighted least-squares system whose minimum-norm solution fixes the common time

Table 16. Public-recording results. Values are reference-packet F1. Rx counts clusters with repeated receivers. Within each scorer, all methods use the same observations, scores, gates, and capture windows. We pool pair counts before calculating F1. We do not average rounded scores.

Evidence	Decoder	3 Aug. F1	2 Sep. F1	Pooled F1	Rx
Timing	Components	0.9972	0.9922	0.9954	1
Timing	Assignment	0.9983	0.9922	0.9961	0
Timing	Constrained	0.9983	0.9922	0.9961	0
Metadata	Components	0.9983	0.9961	0.9975	0
Metadata	Assignment	0.9983	0.9961	0.9975	0
Metadata	Constrained	0.9983	0.9961	0.9975	0

origin; unsupported corrections remain zero. The supported pair median is 0.999985 s, without test-identity fitting.

Corrected time is $t'_i = t_i - \widehat{b}_{s_i}$, with tolerance $G = \max(1\,\text{ms}, Q_{0.99}) = 3.02517\,\text{ms}$. Here $Q_{0.99}$ is the 99th percentile of absolute calibration-reference residuals. This empirical tolerance does not carry the Gaussian guarantee of Proposition 1. The features are

$$\psi_{ij} = \left[\log\left(1 + \frac{|t'_i - t'_j|}{G}\right),\ \mathbf{1}[F_i \neq F_j],\ \mathbf{1}[L_i \neq L_j]\right]. \tag{29}$$

Timing scoring uses the first coordinate; metadata scoring uses all three. Inputs are standardized and logistic coefficients fitted on training data, with temperature fitted on separate calibration data. Admission requires $|t'_i - t'_j| \leq G$.

Reported channel frequency is not oscillator offset, and decoded spreading factor is not a probabilistic recognition output. We synthesize no geometry, RF uncertainty, or class posterior.

Corrected timestamps define fixed 60 s windows without identity-based reassignment. Each scorer uses connected components, pairwise assignment followed by union, and constrained decoding.

Evaluation retains singletons and counts window-split reference pairs as false negatives over each complete capture. We report captures separately and pool TP, FP, and FN. Two captures do not support deployment-population intervals based on individual pairs.

4.13.3. Measured outcomes and interpretation

The 23,677 reports contain 22,260 reference packets, including 1417 observed at both gateways. Metadata constraints produce 1411 true-positive, one false-positive, and six false-negative pairs: precision 0.9993, recall 0.9958, and F1 0.9975. On 3 August, TP is 900, FP 0, and FN 3 (F1 0.9983); on 2 September, TP is 511, FP 1, and FN 3 (F1 0.9961). Assignment and components give the same partition.

Timing-only components produce seven false positives and one receiver conflict. Assignment and constraints each produce five false positives with F1 0.9961 (Table 16).

Shared decoded meanings support a reproducible metadata-to-packet procedure, but metadata already resolves nearly all ambiguity. The two-gateway check has no measured class posteriors or source-specific protocol vocabulary and offers little room to distinguish decoder accuracy.

Assignment is already receiver-exclusive with two gateways, leaving higher-order grouping untested. Metadata F1 0.9975 does not validate multi-protocol robustness or RF uncertainty. Section 4.14 tests the supervised preprocessing behind this result.

Table 17. Public preprocessing sensitivity with unchanged test reports. Train positives count reference pairs available to scorer fitting. Clock corrections require at least ten matches for a receiver pair. G is the empirical time gate. F1 is reference-packet F1. NE means the required fit cannot be estimated.

Preprocessing setting	Train positives	G (ms)	Aug. F1	Sep. F1	Pooled F1
Original 3/1 split	257	3.025	0.9983	0.9961	0.9975
2019 training blocks only	1	1003.010	0.0000	0.0000	0.0000
May training only	256	3.025	0.9983	0.9961	0.9975
April training only	0	–	NE	NE	NE
March train / April calibrate	–	–	NE	NE	NE
2019 train / May calibrate	1	2000.887	0.0000	0.0000	0.0000
Clock correction disabled	257	1003.010	0.9603	0.9202	0.9457
Calibration quantile 0.95	257	1.000	0.9956	0.9892	0.9933
Calibration quantile 0.999	257	3307.621	0.9923	0.9875	0.9905

4.14. Sensitivity of the auxiliary public check

The public sensitivity audit fixes all 23,677 test IDs while varying training, calibration, clock correction, or quantile (Table 17). Original F1 is 0.9975, unchanged with May 2020 training alone. The two 2019 blocks provide one positive pair and no clock pair meeting ten-reference support, giving zero F1. Two settings lack usable training classes or calibration matches and are reported NE.

Without clock correction, metadata F1 is 0.9457. Quantile 0.95 gives F1 0.9933; quantile 0.999 expands the gate to 3.31 s and gives 0.9905. More candidates need not improve association. The original 0.99 rule stays fixed without test-F1 selection.

4.15. Public-reference tails and clock supervision

Of July's 434 reference pairs, four have corrected differences above two seconds despite matching payloads and counters; their reported channels differ. Five May pairs and two September pairs show the same pattern, while all 2096 other 2020 pairs have matching channels. Original CSV checks confirm these fields. A single timestamp without an independent transmission identifier cannot distinguish repetitions from metadata errors; all pairs remain evaluated.

July's 0.99 quantile interpolates between 2.120 and 3.471 ms; the 0.999 quantile lies between 2.136977 and 4.201604 s, explaining the jump to 3.308 s. Channel disagreement further limits the payload reference's physical meaning, precluding interpretation as clock drift alone.

A reference-free training lag histogram uses time, channel, spreading factor, and gateway ID with fixed support and peak-separation rules. Refinement contains 251 candidates with median 0.999985 s, matching the supervised offset numerically. Both fits give identical partitions on 23,677 reports, with TP 1411, FP 1, FN 6, and F1 0.9975. Population selection, scoring, and gate calibration remain reference-assisted (Supplementary Section S15).

The 2019-only fit has one positive pair from gateways absent in the 2020 tests, and April has no cross-gateway reference pair. Zero F1 combines sparse supervision and unsupported calibration rather than isolating drift. These retrospective checks diagnose preprocessing and reference construction.

5. Discussion

5.1. What the semantic result establishes

The method registers report meanings before grouping events. Field definitions and the common vocabulary are supplied, while missing permutations can be learned from provisional groups. New concepts, many-to-one mappings, and hierarchies require additional models.

Relative compatibility and absolute names remain distinct: common renaming preserves dot products and event groups. The known reference fixes names for hidden-entry evaluation, while event F1 evaluates grouped reports. Neither metric replaces the other.

Pair tables provide exact conditional optima for small vocabularies and avoid coordinate traps, but do not establish general superiority. Exact-minus-spectral intervals include zero, and forty scenes also allow reference-only recovery. Event groups remain approximate. Sequential and alternating maps coincide in all 150 nominal and same-class validation settings, supporting recovery from physical groups without benefit from repeated event updates.

Complete registration still leaves dense same-class F1 near 0.45, while shared wrong predictions can preserve event compatibility. Mapping accuracy, class correctness, and event F1 therefore measure different properties and need separate evaluation.

5.2. Feasibility, accuracy, and dependence

Feasibility is constructed, not evidence of correct identity. Physical-only constraints lose 0.0332 F1 against assignment with interval $[-0.0439, -0.0225]$, whereas boosted constraints gain 0.0108 with $[0.0020, 0.0196]$. These scorer-dependent costs motivate reporting F1, ARI, B^3 F1, exact recovery, and violations together.

Concentration does not measure correctness. Unweighted scoring improves the nominal reference, whereas imbalance and receiver-specific errors reduce semantic gains. Event-shared errors can preserve agreement. Shared receiver errors also make J composite, so pair calibration provides no calibrated group posterior. Replication intervals describe generator variation rather than group confidence.

The Gaussian bound requires a correct zero-mean error model and can fail under excess delay. Fragmentation conflicts with exclusivity; correlated false detections can form feasible wrong groups. Consolidation must identify repeated detections and propagate uncertainty. Shared artifacts require joint false-alarm modeling or nuisance-event evidence because lowering confidence alone cannot identify them. These unfitted models require measured detections and independent event labels. Pairwise admission still does not prove a common source position.

5.3. Measured evidence and remaining validation

LoED's conservative exclusions select a restricted traffic population. Equal payloads heard by complementary gateways can still arise from different transmissions, with no independent instrument resolving identity. The high F1 is conditional on this population and supervised preprocessing.

Only two of the original nine gateways remain eligible. Decoded-SF swaps supply no measured posteriors or higher-order semantic validation, while channel-discordant equal-payload pairs remain ambiguous. Training-metadata clock fitting removes one source of supervision, but population selection, scoring, and gate calibration still use references.

Field validation needs synchronized recordings from at least three receivers, independent event labels, documented native vocabularies, and measured class posteriors. WiSig addresses another identity task [26]; the Sorbonne resource is a potential future source [50,51], but suitable annotations remain to be established. No unverified substitute is evaluated.

6. Conclusions

We formulate mapping-assisted semantic alignment for RF receiver reports. Known field meanings support physical comparison, and mapped class probabilities support event

association. Receiver exclusivity and physical admission restrict the output partition, with experiments evaluating these feasibility requirements separately from event accuracy.

The mapping experiments hide three receiver maps while retaining a known reference to fix common class names. Independent adaptation scenes recover missing correspondences within the four-class vocabulary. Exact conditional search avoids a coordinate-update failure, while spectral synchronization provides comparable accuracy in the tested conditions. Alternating updates do not improve on sequential estimation. Wrong relative mappings harm association, whereas a shared class renaming leaves it unchanged.

Dense same-class collisions remain difficult after complete map recovery because correct column meanings cannot distinguish every transmission. Shared wrong predictions can also preserve event compatibility, separating semantic registration from class correctness and event identity. Integer programs quantify greedy objective loss and report-processing cost. LoED remains an auxiliary metadata check. Its timestamp audit exposes packet-reference ambiguity. A training-metadata clock estimate removes one use of reference identities in that check. Field validation still needs independent event truth and measured class probabilities from several receivers.

Supplementary Materials: Supporting information is provided in the separately submitted file `supplement.pdf`: Sections S1–S15 (generator and implementation details, metric definitions, proofs, extended validation, and reproduction protocols); Figures S1–S6 (selective pair risk, exact-audit objective loss, the observation interface, evidence features, calibration, and semantic stress); and Tables S1–S16 (observation schemas, stress settings, partition metrics, notation, public-data inclusion, complete mapping and decoder results, pair counts, computational costs, uncertainty intervals, timestamp tails, clock fitting, and report-path latency).

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Data Availability Statement: The experimental code, data, saved models, and results supporting this study are available at https://github.com/XueT-BZ/Physics-semantic-rf-association. Data provenance, licensing information, and reproduction instructions are included in the repository. The public LoED sample retains its original attribution and license. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: Yuanfeng Gao is currently pursuing the Ph.D. degree with Beijing University of Posts and Telecommunications, Beijing, China. He is currently a Professor-level Senior Researcher with the 54th Research Institute, China Electronics Technology Group Corporation (CETC) where Bojun Zhao is also employed. Wenbo Xu is a professor at Beijing University of Posts and Telecommunications. Apart from these disclosed affiliations and employment relationships, the authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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